Shubham Kumar

M.Tech Thesis

M.Tech Thesis
M.Tech Student
National Institute of Technology, Patna

Document Details

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

- **105** Not Cited or Quoted 8%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

3% Internet sources
7% Publications
4% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **105** Not Cited or Quoted 8%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

3% Internet sources
7% Publications
4% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Publication	**Abdulla Juwaied, Lidia Jackowska-Strumillo. "Applying Machine Learning to DEEC ...**	<1%
2	Student papers	**Maharishi University Of Information Technology, Lucknow**	<1%
3	Student papers	**Georgia Institute of Technology Main Campus**	<1%
4	Student papers	**Universite De Tlemcen**	<1%
5	Student papers	**Anna University**	<1%
6	Student papers	**National Institute of Technology Jamshedpur**	<1%
7	Publication	**Raed Alsaqour, Elmustafa Sayed Ali, Rania A. Mokhtar, Rashid A. Saeed, Hesham ...**	<1%
8	Internet	**www.geeksforgeeks.org**	<1%
9	Publication	**Andrew Houston, Georgina Cosma. "A meta-heuristic approach to estimate and e...**	<1%
10	Publication	**Chaima Derouiche, Abdallah Bensayah, Khadra Bouanane, Oussama Aiadi, Belal ...**	<1%

11 Publication

Tripathy, Asis, and Suchismita Chinara. "Evolution of Virtual Clustering in Wireles... <1%

12 Publication

Chowdhury, Areefin-Ul-Hassan. "Mitigating the Greenhouse Gas Footprint of Hyd... <1%

13 Publication

Vikram Kumar, Santos Kumar Das. "Physical layer impairment-aware ant colony o... <1%

14 Student papers

University of Mosul <1%

15 Publication

Thet Htar Su, Shaswot Shresthamali, Masaaki Kondo. "Quantum framework for re... <1%

16 Publication

Deepak Sharma, Amritesh Ojha, Amol P. Bhondekar. "Heterogeneity consideratio... <1%

17 Publication

Enan A. Khalil, Bara'a A. Attea. "Stable-Aware Evolutionary Routing Protocol for W... <1%

18 Publication

Mohammed ALSULTAN, Kasim OZTOPRAK, Reza HASSANPOUR. "Power Aware Ro... <1%

19 Internet

www.mdpi.com <1%

20 Publication

Abdulla Juwaied, Lidia Jackowska-Strumillo, Artur Sierszeń. "Enhancing Clustering... <1%

21 Publication

Arvind Dagur, Sohit Agarwal, Dhirendra Kumar Shukla, Shabir Ali, Sandhya Sharm... <1%

22 Publication

Fuad Bajaber, Irfan Awan. "Energy efficient clustering protocol to enhance lifetim... <1%

23 Student papers

University of Edinburgh <1%

24 Publication

Kiran Singh, Suman Avdhesh Yadav, T. Poongodi. "Efficient multitasking in hetero... <1%

25 Publication
Krishan Arora, Himanshu Sharma, Dwarkadas Prahladadas Kothari. "Artificial Int... <1%

26 Publication
Piyush Rawat, Siddhartha Chauhan. "Clustering protocols in wireless sensor netw... <1%

27 Publication
Warrier, A.. "How much energy saving does topology control offer for wireless se... <1%

28 Internet
www.researchgate.net <1%

29 Publication
K. V. Sambasivarao, Anasuya Sesha Roopa Devi Bhima. "Artificial Intelligence, Co... <1%

30 Student papers
University College London <1%

31 Student papers
Chandigarh Group of Colleges <1%

32 Internet
nrl.northumbria.ac.uk <1%

33 Internet
www.grin.com <1%

34 Student papers
Gitam University <1%

35 Student papers
University of Technology <1%

36 Internet
oulurepo.oulu.fi <1%

37 Publication
Paolo Ferro, Harinadh Vemanaboina, Chander Prakash. "Computational Techniqu... <1%

38 Internet
cgu-odisha.ac.in <1%

39 Publication

Ademola Philip Abidoye, Boniface Kabaso. "Energy-efficient hierarchical routing i... <1%

40 Student papers

Indian School of Mines <1%

41 Student papers

Northcentral <1%

42 Publication

Nour El Houda Bourebia, Soumia Bourebia. "RMIS: an independent vertex set bas... <1%

43 Student papers

Royal Holloway and Bedford New College <1%

44 Student papers

Sri Sairam Engineering College <1%

45 Internet

doaj.org <1%

46 Student papers

Indian Institute of Technology, Bombay <1%

47 Student papers

Liverpool John Moores University <1%

48 Internet

www.ijsrd.com <1%

49 Student papers

Melbourne Institute of Technology <1%

50 Student papers

University of Anbar <1%

51 Internet

cris.vtt.fi <1%

52 Internet

ijirt.org <1%

53 Publication

Azzedine Boukerche, Lining Zheng, Omar Alfandi. "Outlier Detection", ACM Comp... <1%

54 Publication

Mümün Çalışkan, Murtaza Cicioğlu. "Comparative analysis and sensitivity evaluat... <1%

55 Publication

Sinan Q. Salih, AbdulRahman A. Alsewari, H. A. Wahab, Mustafa K. A. Mohammed,... <1%

56 Publication

Soro, S.. "Cluster head election techniques for coverage preservation in wireless ... <1%

57 Internet

academia.wascal.org <1%

58 Internet

es.slideshare.net <1%

59 Internet

jds-online.org <1%

60 Internet

pure-oai.bham.ac.uk <1%

61 Internet

www.sysbiosci.com <1%

31
5
22

Chapter 1

Introduction

1.1 General

Wireless Sensor Networks (WSNs) have emerged as a critical enabling technology for real time monitoring and automation in harsh and safety critical environments. A WSN generally made up of a large number of low power sensor nodes and these nodes are used for sensing, processing and transmitting environmental data to the central sink or base station. Due to their self organizing nature, scalability and ease of deployment, WSNs have found widespread application in industrial automation, environmental surveillance, disaster management, healthcare monitoring and military systems. Among these applications, underground mining represents one of the most challenging operational domains for WSN deployment. Underground mines are characterized by narrow and irregular tunnel geometries, multi-level galleries, sharp turns, metallic obstructions and highly heterogeneous propagation conditions. Tunnel lengths can extend from 2 km to over 5 km, with working depths reaching up to 4 km below the surface. These physical constraints result in severe signal attenuation, multi-path fading, shadowing and frequent link instability, making reliable wireless communication extremely difficult. In underground mines, sensor nodes are deployed to continuously monitor critical safety parameters such as methane and carbon monoxide concentrations, temperature, humidity, vibration levels and structural stability indicators. Any failure or delay in transmitting this information can lead to catastrophic consequences, including explosions, suffocation and structural collapses. However, these tiny nodes are powered by small, non rechargeable batteries and replacing or recharging these batteries in hazardous underground conditions is complex, expensive and often infeasible. Due to this, energy efficiency becomes the dominant design constraint in underground WSNs. Conventional routing and clustering protocols, originally designed for terrestrial or open field deployments, fail to perform optimally in such constrained environments. While clustering based protocols reduce transmission overhead by aggregating data at Cluster Heads (CHs), they often suffer from uneven energy consumption. CHs, which handle additional communication and processing tasks, tend to exhaust their energy much faster than ordinary nodes, leading to the formation of energy holes and early network partitioning.

21

To mitigate these challenges, this research introduces a novel framework termed Reinforcement Learning with Hybrid Clustering (RLHC). The proposed framework integrates learning based intelligence with multi layer clustering mechanisms to enable adaptive, energy aware decision making. By allowing the network to learn from its operational experience and dynamically optimize cluster formation and routing behavior, RLHC aims to significantly enhance network lifetime, reliability and throughput in underground mining environments. This approach moves beyond static heuristics and introduces a self optimizing network paradigm capable of responding to environmental dynamics and energy variations over time.

1.2 Project background

The mining industry plays a vital role in India's economic and industrial development, contributing significantly to energy production, infrastructure growth and raw material supply. Major mining operations are concentrated in states such as Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh and Karnataka, encompassing both metallic and non-metallic mines. Despite technological advancements, underground mining remains one of the most hazardous industrial activities worldwide. According to data reported by the Directorate General of Mines Safety (DGMS), between 2010 and 2025, more than 20,025 mining accidents were recorded in India, resulting in approximately 49,615 fatalities. A detailed analysis of these incidents reveals that nearly 40% of fatalities are caused by toxic gas leaks, particularly methane and carbon monoxide, while 35% result from roof collapses. The remaining accidents are attributed to explosions, equipment malfunctions and ventilation failures. A recurring factor in many of these incidents is the failure of sensing or communication infrastructure. In several cases, sensor nodes responsible for gas detection or structural monitoring cease operation due to energy depletion, leaving critical areas unmonitored. As a result, hazardous conditions remain undetected until they escalate into fatal events. Continuous and reliable monitoring is therefore essential to ensure early warning and timely evacuation.

Current underground monitoring systems often rely on static communication protocols that are not designed to handle the energy constraints, dynamic conditions and harsh propagation characteristics of mines. High-frequency data transmission, required for real time safety monitoring, rapidly drains node batteries and shortens network lifetime. This project is motivated by the urgent need to modernize underground mine monitoring systems by introducing intelligent, adaptive communication frameworks. By extending network lifetime and maintaining consistent sensing coverage, the proposed RLHC framework seeks to enhance the reliability of underground safety systems. The ultimate goal is to enable continuous, uninterrupted monitoring, thereby reducing the risk of undetected hazards and contributing to the prevention of large-scale loss of life in underground mining operations.

1.3 Need of the study

Despite of in-depth research on WSNs clustering and routing protocols, existing solutions such as LEACH and DEEC exhibit several limitations when applied to underground mining environments. These protocols rely on probabilistic or heuristic based CH selection mechanisms, which do not adequately consider node heterogeneity or real time network conditions. In heterogeneous WSNs, nodes often possess different initial energy levels and experience varying energy depletion rates. However, traditional protocols may still select low energy nodes as CHs, causing their premature failure. This results in coverage gaps or "blind spots" that compromise the safety monitoring system. Furthermore, these protocols assume relatively stable communication links and static deployments, assumptions that do not hold in underground mines where environmental conditions and node positions may change frequently. Underground environments are inherently dynamic. Sensors may be attached to moving machinery or personnel and atmospheric conditions such as humidity, dust concentration and gas density can significantly affect wireless link quality. Static protocols lack the ability to adapt to these changes, leading to frequent packet losses and inefficient energy usage. The need for this study arises from the requirement for an intelligent, adaptive decision making architecture capable of responding to real time network feedback. Reinforcement Learning, particularly Q-Learning, offers a powerful mechanism for such adaptation. Through continuous interaction with the environment, an RL agent can evaluate the long term impact of its actions such as selecting a CH, adjusting transmission power, or rerouting data and converge toward an optimal policy that balances energy consumption and communication reliability. By embedding RL within the clustering and routing process, the proposed RLHC framework enables the network to self optimize and dynamically reconfigure itself. This adaptive capability is essential for maintaining long term network stability and ensuring reliable safety monitoring in underground mining WSNs.

1.4 Scope of the study

58 The scope of this research is focused on the design, implementation and performance evaluation of the RLHC framework specifically tailored for underground mining environments. The proposed framework adopts a five layer hierarchical architecture, each layer addressing a distinct aspect of energy efficiency and communication reliability. DEEC layer identifies potential CH candidates based on residual energy, ensuring that high energy nodes are preferentially selected for energy intensive roles. Energy Efficient Knapsack Algorithm (EEKA) layer optimizes CH selection by ensuring spatial diversity and centrality, thereby reducing communication bottlenecks and load imbalance. K-Means clustering layer groups sensor nodes into compact clusters to minimize intra cluster transmission distance and energy consumption. Reinforcement Learning layer employs a Q-learning agent that observes network states such as residual energy, Packet Delivery Ratio (PDR) and node density to make adaptive decisions. Communication management layer implements multi-hop routing using Dijkstra's algorithm under a composite channel model incorporating Rician and Log-Normal fading effects.

The evaluation of the proposed framework is conducted exclusively through simulation based experiments using Python 3.11.1. A network consisting of 150
51 sensor nodes deployed over a 1000 m × 1000 m area is modeled. The performance of RLHC, including both Centralized RLHC and Federated RLHC, is compared against conventional clustering protocols such as LEACH and DEEC and meta heuristic approaches such as PSO, ACO and FCM. Key performance metrics analyzed include network lifetime, sector-wise survivability, throughput, packet delivery ratio and total energy consumption per round. While the study does not include physical hardware implementation, it establishes a strong theoretical and algorithmic
36 foundation for future real world deployment.

1.5 Organization of thesis

This thesis is organized into seven chapters, structured to systematically present the research problem, proposed methodology, experimental evaluation and final conclusions. Chapter 1 that is Introduction, this chapter introduces the research domain of underground WSNs and highlights the safety critical challenges associated with energy efficiency, reliability and network longevity in mining environments. It presents the motivation for the study, discusses the limitations of existing protocols, outlines the need for intelligent and adaptive energy optimization and defines the scope and objectives of the proposed RLHC framework. Chapter 2 is literature review, this chapter provides a comprehensive review of existing research related to clustering and routing protocols in WSNs, including classical approaches, meta heuristic techniques, reinforcement learning based solutions and federated learning frameworks. The chapter critically analyzes prior work and identifies key research gaps related to adaptability, energy balance, scalability and reliability in underground mining scenarios. Chapter 3 represents problem statement and system model which formally defines the research problem addressed in this thesis. It presents the system model for underground mining WSNs, including network assumptions, energy consumption models, node heterogeneity, sector-based deployment and communication constraints. Chapter 4 focus on proposed RLHC framework and theoretical foundations which describes the proposed solution in detail. It first explains the theoretical foundations, including the mathematical models for energy consumption, network heterogeneity, Markov Decision Processes (MDP) and Bellman equations governing reinforcement learning. The chapter then presents the design and operation of the proposed five layer RLHC architecture, detailing the DEEC, EEKA, K-Means, Q-learning based decision making and communication management strategies. Chapter 5 is simulation setup and performance evaluation, this chapter outlines the simulation environment, parameter configurations and evaluation metrics used to assess the performance of the proposed framework. It presents quantitative results for network lifetime, energy consumption, throughput, pdr and sector-wise survivability, comparing RLHC (Centralized and Federated) with baseline and meta heuristic protocols. Chapter 6 describe analysis and discussion and provides an in-depth analysis and interpretation of the simulation results presented in Chapter 5. It discusses the observed performance trends, explains the reasons behind RLHC's improvements in energy efficiency and reliability and evaluates the trade offs between centralized and federated learning approaches. The implications of sector-wise survivability, scalability and robustness for safety critical underground mining applications are critically examined. Chapter 7 focus on conclusion and future directions which summarizes the major contributions and findings of the research. It highlights the practical significance of the proposed RLHC framework for underground mine monitoring and outlines potential directions for future work, including multi tier federated RLHC architectures, emergency aware and fault-tolerant policies, real world deployment challenges and integration with edge intelligence platforms.

Chapter 2

Literature Review

WSNs have emerged as a core enabling technology for modern Internet of Things (IoT) ecosystems, supporting applications across healthcare, agriculture, industrial automation, environmental monitoring and underground mining. Despite their wide applicability, WSNs continue to face critical challenges related to energy efficiency, network stability, scalability and reliability. Since sensor nodes are typically battery powered and often deployed in harsh environments, efficient energy management directly determines network lifetime and overall performance. Over the past two decades, researchers have proposed a wide range of clustering protocols, optimization techniques and intelligent learning based strategies to address these challenges. This chapter discuss a detailed review of existing literature related to energy efficient clustering and routing in WSNs, with a particular focus on classical protocols, heterogeneity aware designs, meta heuristic optimization methods, fuzzy logic based approaches, reinforcement learning techniques and hybrid frameworks. The limitations of existing solutions are critically analyzed to motivate the proposed RLHC framework.

2.1 Classical Clustering Protocols

Clustering is widely recognized as one of the most effective techniques for improving energy efficiency in WSNs. By organizing sensor nodes into clusters and assigning CHs to aggregate and forward data to the BS, clustering significantly reduces redundant transmissions and communication overhead

2.1.1 LEACH and Its Variants

The LEACH protocol is among the earliest and most influential clustering protocols proposed for WSNs. LEACH introduces random rotation of CHs to distribute energy consumption evenly among nodes and reduce direct long range communication with the BS [10]. By periodically re-electing CHs, LEACH improves network lifetime compared to flat routing schemes. However, LEACH suffers from many well known limitations, including, random and uneven CH distribution, lack of residual energy awareness poor performance in heterogeneous and large-scale networks. These shortcomings have led to the development of numerous LEACH variants that incorporate residual energy, distance, or node density into CH selection. The core equations that guide LEACH is expressed as following:

$$T(i) = \begin{cases} \frac{P_i}{1 - P_i \cdot \left(r \bmod \left(\frac{1}{P_i}\right)\right)}, & \text{if } i \in G \\ 0, & \text{otherwise} \end{cases}$$

Where i^{th} node probability is *Pi* which get selected as the CH and *T(i)* represents the cluster head election threshold function. The LEACH is having multiple rounds and each round consists of two distinct phases such as the setup phase and the steady state phase. During the setup phase, clusters are formed and CHs are elected. Each node independently determines whether it should become a CH for the current round using a probabilistic threshold function. Nodes that elect themselves as CHs broadcast an advertisement message to neighboring nodes. Non CH nodes then select the CH with the strongest received signal strength and inform the chosen CH of their membership. Once cluster membership is established, the CH creates a TDMA schedule for intra-cluster communication. In the steady-state phase, sensor nodes transmit sensed data to their respective CHs according to the TDMA schedule. The CH collect the received data to eliminate redundancy and relay the collected data to the BS. Since the steady-state phase is significantly longer than the setup phase, the overhead associated with clustering is amortized over multiple data transmission cycles. After that the randomized CH rotation happens which is one of the most important innovations introduced by LEACH is randomized rotation of CHs. The CH role is energy-intensive due to the additional burden of data aggregation and long-range transmission to the BS. To prevent early death of CH nodes, LEACH ensures that the CH responsibility is periodically rotated among all nodes in the network. This rotation mechanism promotes load balancing and ensures that energy consumption is distributed as evenly as possible across the network. Over time, each node has an approximately equal probability of serving as a CH, which enhances network stability and prolongs network lifetime during the initial operational phase.

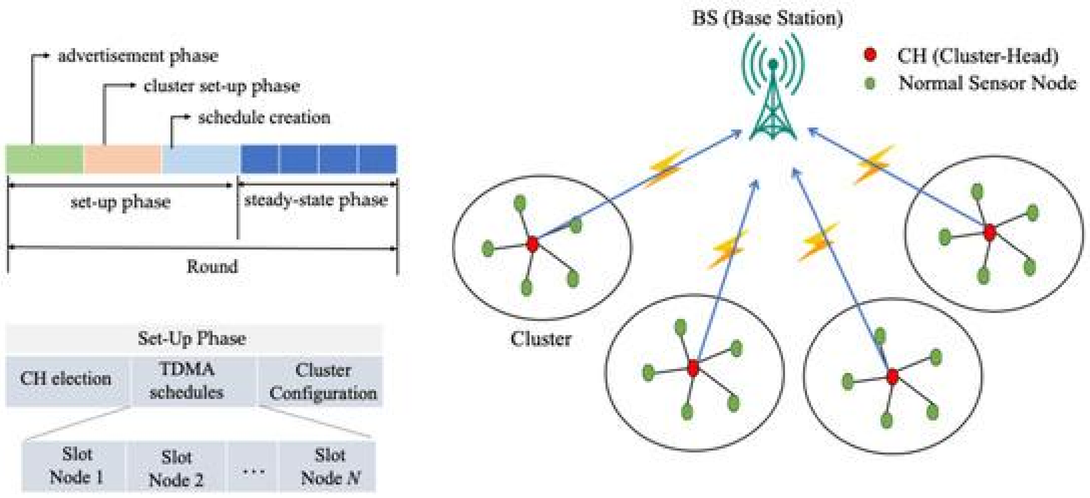


Figure 2.1.1 Representation of the LEACH WSNs for data gathering [5].

Performance Advantages of LEACH, LEACH offers several advantages that contributed to its widespread adoption and influence, energy efficiency by reducing direct transmissions to the BS and enabling data aggregation, LEACH significantly lowers energy consumption compared to flat routing protocols. Distributed operation CH selection does not require global network knowledge, improving scalability. Load balancing by randomized CH rotation distributes energy consumption across nodes and simplicity by making it easy to implement and computationally lightweight. These advantages make LEACH particularly suitable for small scale, homogeneous WSN deployments with static node positions.

Despite its pioneering contributions, it exhibits several critical limitations that restrict its applicability in real world scenarios such as random and uneven CH distribution.

The probabilistic CH election process can lead to uneven spatial distribution of CHs, resulting in imbalanced clusters. Also the lack of residual energy awareness, it does not consider residual energy during CH selection, allowing low energy nodes to become CHs. Homogeneous network assumption and assumes equal initial energy across nodes, making it unsuitable for heterogeneous WSNs. Single hop communication does direct CH-to-BS communication limits scalability in large scale networks. Poor adaptability which does not respond dynamically to changing network conditions such as node failures or energy depletion patterns. These limitations are particularly severe in underground mining WSNs, where communication conditions are harsh, node replacement is impractical and energy efficiency is paramount.

2.1.2 DEEC and Energy Aware Extensions

The DEEC protocol is a well established extension of the LEACH protocol that addresses one of its most critical limitations that is the lack of residual energy awareness during CH selection. While LEACH assumes homogeneous energy distribution and assigns equal probability for all nodes to become CHs, DEEC introduces an adaptive probability mechanism based on each node's residual energy relative to the network's average energy. This enhancement enables more balanced energy consumption and significantly extends network lifetime, particularly in heterogeneous WSNs[11]. DEEC was specifically designed to operate in multi-level heterogeneous networks, where sensor nodes may have different initial energy levels. Such heterogeneity is common in real world deployments due to hardware variation, partial node replacement, or role specific energy provisioning. By incorporating residual energy into the CH election process, DEEC ensures that energy rich nodes have a greater share of the communication burden, while low energy nodes are protected from premature depletion. It introduce the principle of energy aware CH selection. The core idea behind DEEC is that CH selection probability should be proportional to the ratio of a node's residual energy to the average energy of the network. This contrasts sharply with LEACH's purely probabilistic approach, which does not differentiate between nodes based on their energy states. In DEEC, each node independently computes its probability of becoming a CH using the following expression:

$$P_i = prop \times \left(\frac{E_i}{\bar{E}(r)} \right)$$

where *Pi* is the probability of node *i* becoming a CH in round *r*, *Ei(r)* is the residual energy of node *i* at round *r* and *Eavg(r)* is the average residual energy of all nodes in the network at round *r*. This formulation ensures that nodes with higher residual energy are more likely to be selected as CHs, while nodes with depleted energy reserves are less frequently assigned energy intensive roles.

Similar to LEACH, the DEEC protocol employs a threshold based mechanism to determine and finalize CH selection in each round. In DEEC, this threshold function is designed to incorporate the residual energy of individual nodes relative to the average energy of the network, thereby enabling energy-aware and adaptive CH election in heterogeneous WSN environments. Unlike LEACH, which assigns equal

CH selection probability to all nodes, DEEC dynamically adjusts the CH probability so that nodes with higher remaining energy are more likely to become CHs. The threshold function in DEEC is mathematically defined to regulate CH candidacy within a given round and ensures that each node becomes a CH approximately once every epoch, proportional to its energy level. By comparing a randomly generated number with the threshold value, a node autonomously decides whether to assume the CH role. This distributed decision-making process eliminates the need for centralized coordination while promoting balanced energy consumption across the network. The threshold function is expressed as

$$T(i) = \begin{cases} \frac{P_i}{1 - P_i \cdot \left(r \bmod \left(\frac{1}{P_i}\right)\right)}, & \text{if } i \in G \\ 0, & \text{otherwise} \end{cases}$$

where *T(i)* represents threshold for node *i* to become CH, *Pi* is probability of node *i* becoming a CH, *r* is current round number and *G* is set of nodes that have not been CHs in the last *1/Pi* rounds. DEEC naturally accommodates this heterogeneity without requiring explicit role classification. Since advanced nodes maintain higher residual energy for longer duration, they are automatically selected as CHs more frequently. This dynamic adaptation improves stability period, delays the first node death (FND) and extends the overall network lifetime. Subsequent DEEC variants such as DDEEC, EDEEC and TDEEC have further refined this approach by adjusting probability thresholds to prevent over penalization of high energy nodes, which may otherwise suffer from excessive CH selection. By aligning CH selection with residual energy, DEEC achieves superior energy balancing compared to LEACH. Various studies consistently demonstrate that DEEC extends the stability period (time until first node death), delays network partitioning, improves throughput and packet delivery reliability. These properties make DEEC particularly attractive for applications where continuous monitoring is critical, such as environmental surveillance, industrial automation and underground mine safety systems. Despite its advantages,

But DEEC still suffers from several inherent limitations that restrict its effectiveness in highly dynamic or large-scale deployments that is DEEC relies on predefined probability equations that do not adapt to rapid changes in network topology, traffic load, or channel conditions, also nodes with higher initial energy may be selected as CHs too frequently, leading to early exhaustion if probability scaling is not carefully controlled. DEEC considers only residual energy, ignoring critical metrics such as link quality, pdr, node mobility, or sector-wise survivability. The protocol does not learn from past decisions or optimize CH selection based on long-term outcomes. Energy efficiency is treated as the sole optimization goal, whereas real world deployments require multi objective trade-offs between energy, reliability, latency and coverage. These limitations are especially problematic in underground mining environments, where communication links are unstable, node accessibility is limited and failures can have catastrophic consequences.

2.1.3 Early Reinforcement Learning Based LEACH Enhancements

In recent years, RL is popular as a promising paradigm for improving the adaptability and intelligence of clustering protocols in WSNs. Unlike classical heuristic based approaches, RL enables sensor nodes or CHs to learn optimal decision making policies through continuous interaction with the network environment. Several studies have integrated RL with the LEACH protocol to overcome its static and probabilistic nature, resulting in adaptive clustering mechanisms capable of responding to dynamic network conditions [12]. RL enhanced LEACH variants aim to replace or augment the traditional random CH selection process with a learning based decision model. In these approaches, nodes are treated as agents that observe network states such as residual energy, node density, distance to the base station and historical communication success and select actions that maximize a long term reward. The reward function is typically designed to reflect key performance objectives, including energy efficiency, throughput, stability period and network lifetime. Protocols such as LEACH-RL and ReLeC (Reinforcement Learning enhanced Clustering) employ RL techniques, most commonly Q-learning, to dynamically determine CH roles. Instead of relying solely on probabilistic thresholds, these protocols allow nodes to evaluate the long-term consequences of becoming a CH by associating rewards or penalties with their decisions. In LEACH-RL, each node maintains a Q-table that maps observed states to actions such as electing itself as a CH, remaining a normal member node, adjusting transmission power or cluster affiliation. Over successive rounds, nodes update their Q-values using feedback from the environment, enabling the network to gradually converge toward energy-efficient CH selection patterns. ReLeC further enhances this process by incorporating additional state parameters such as local node density and inter node distances, which helps reduce cluster imbalance and improves spatial distribution of CHs. Simulation based evaluations reported in the literature consistently demonstrate that RL-enhanced LEACH protocols outperform classical LEACH in several key metrics. By avoiding repeated selection of low energy nodes as CHs, RL based protocols reduce premature node failures and balance energy consumption more effectively. The time until the FND is significantly increased due to intelligent CH role assignment. Better cluster formation and adaptive routing decisions result in increased packet delivery to the BS. Learning based decisions reduce cluster instability and frequent reconfiguration, improving overall network robustness. These benefits highlight the potential of RL to address the rigidity of traditional clustering protocols and enable self-optimizing WSNs capable of adapting to evolving conditions.

Despite their advantages, RL based LEACH variants introduce notable computational and communication overhead, which poses a significant challenge for resource constrained IoT and sensor nodes. Maintaining and updating Q-tables requires additional memory and processing power, which may not be feasible for low cost sensor hardware. Furthermore, RL-based protocols often rely on state information exchange among nodes or between nodes and a centralized controller. This exchange increases control packet overhead and consumes additional energy, partially offsetting the gains achieved through intelligent CH selection. In dense networks, the overhead associated with learning can become substantial, leading to faster battery depletion. Another critical limitation of RL-based LEACH protocols is convergence behavior. Q-learning algorithms typically require a sufficient number of interactions to converge to an optimal policy. In dynamic WSN environments where nodes may fail, move, or experience fluctuating channel conditions the learning process may never fully converge, resulting in sub optimal or unstable clustering decisions. Moreover, many RL-based LEACH implementations assume centralized training or global knowledge of the network state. While this simplifies learning, it severely limits scalability and introduces a single point of failure. Centralized controllers may become communication bottlenecks, particularly in large-scale or geographically dispersed networks such as underground mining environments. RL based LEACH variants also struggle in highly dynamic scenarios involving node mobility, varying traffic patterns, or rapidly changing environmental conditions. Static state definitions and fixed reward functions may fail to capture complex, real-world dynamics. Additionally, exploration mechanisms in RL can temporarily degrade performance, which is unacceptable in safety-critical applications where consistent reliability is mandatory. These limitations underscore the need for lightweight, distributed and energy-aware learning mechanisms that minimize overhead while retaining adaptability.

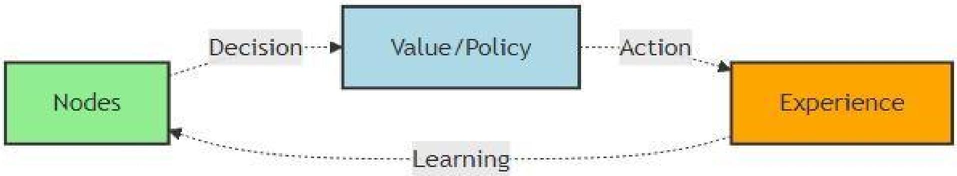


Figure 2.2.3 Generic RL based decision making framework [12]

This figure illustrates the fundamental interaction loop between a learning agent and its environment, which forms the basis of reinforcement learning driven optimization in WSNs. As per literature, this framework consist of agent which observes the current state of the network, such as residual energy, node density, link quality, packet delivery ratio, or traffic load. Based on the observed state, then the RL agent selects an action. Then transitions of environment to a new state happens and provides a reward that quantifies the effectiveness of the chosen action, typically in terms of energy efficiency, throughput, reliability, or network lifetime. The agent uses the reward feedback to update its policy or value function, enabling it to learn an optimal decision making strategy over time. Through repeated interaction, the framework allows the system to adapt dynamically to changing network conditions, balance energy consumption and improve long term performance.

2.2 Heterogeneity Aware Clustering Protocols

To overcome the limitations of homogeneous WSN assumptions, researchers have proposed a range of protocols that explicitly account for node heterogeneity in terms of initial energy levels, hardware capabilities and communication ranges. In practical deployments, sensor nodes are rarely identical; variations arise due to differences in battery capacity, sensing roles, processing power, or incremental node replacement over time. Heterogeneity-aware protocols exploit these differences by assigning energy-intensive tasks, such as cluster head selection or long-range data transmission, to more capable nodes. By incorporating heterogeneity into network design, these protocols achieve improved energy balancing, longer stability periods and enhanced overall network lifetime. Nodes with higher energy reserves or superior communication capabilities are preferentially selected as cluster heads, while low-energy nodes are protected from premature depletion. This approach is particularly beneficial in large-scale and mission-critical applications, such as underground mining and industrial monitoring, where uniform energy consumption and continuous coverage are essential for reliable operation.

2.2.1 SEP Based Heterogeneous Models

The Stable Election Protocol (SEP) is one of the earliest clustering protocols designed specifically for heterogeneous WSNs, where nodes possess different initial energy levels. SEP introduces a weighted CH election mechanism that assigns higher CH selection probabilities to energy rich nodes, referred to as advanced nodes, while normal nodes are assigned lower probabilities. This weighted election ensures that nodes with greater energy reserves assume CH responsibilities more frequently, thereby balancing energy consumption and extending the network's stability period [13]. In a typical two-level heterogeneous SEP model, a fraction *m* of the total nodes are advanced nodes with α times higher energy than the normal nodes. *Eo* denotes the initial energy of a normal node, then the initial energy of an advanced node is given by the expression

$$E_{adv} = E_0(1+\alpha) \qquad\qquad P_{adv} = \frac{p(1+\alpha)}{1+\alpha m}$$

Where the desired percentage of CHs is p, the fraction of advanced nodes is m, α represents the additional energy factor. These equations ensure that the expected number of CHs per round remains constant, while advanced nodes are selected as CHs more frequently than normal nodes. Consequently, SEP significantly improves the stability period, defined as the time until the first node dies. Building upon SEP, the DEEC protocol generalizes heterogeneity support to multi-level heterogeneous networks by dynamically adjusting CH selection probabilities based on residual energy rather than fixed initial energy classes. Unlike SEP, which assigns static probabilities, DEEC continuously adapts to the evolving energy states of the network. Both SEP and DEEC have been shown to significantly enhance network lifetime, stability period and energy efficiency in heterogeneous WSN deployments. By prioritizing energy rich nodes for CH roles, these protocols reduce premature node deaths and maintain more consistent network coverage. However, despite these improvements, SEP and DEEC remain fundamentally heuristic driven and rely on predefined mathematical models. They lack intelligent decision making capabilities and do not adapt to complex network dynamics such as node mobility, fluctuating traffic patterns, varying channel conditions, or sector-wise survivability requirements. Additionally, these protocols optimize primarily for energy efficiency and do not explicitly consider reliability metrics such as packet delivery ratio or latency. These limitations motivate the integration of learning based approaches, such as RL, to enable adaptive, multi objective optimization in heterogeneous and dynamic WSN environments.

2.2.2 Spatial and Centrality Aware Enhancements

Recent enhancements in clustering protocols extend beyond energy awareness by explicitly incorporating spatial characteristics such as node centrality, spatial distribution and inter-node distances into the cluster head (CH) selection process. The motivation behind this approach is that even energy-rich nodes may become inefficient CHs if they are poorly positioned within a cluster, resulting in excessive intra-cluster communication cost and uneven energy dissipation. To address this limitation, algorithms such as the EEKA formulate CH selection as an optimization problem, where the objective is to ensure spatially and centrally distributed CHs which jointly minimize communication energy while satisfying energy and spatial constraints.

In spatially aware clustering approaches, the effectiveness of a CH is determined not only by its residual energy but also by its geometric position relative to member nodes. Two key metrics commonly used to quantify this spatial suitability are the intra-cluster distance cost Dj and the node centrality metric Cj. The intra-cluster distance cost Dj represents the total communication distance between a candidate CH j and all nodes within its cluster. It is defined as the sum of Euclidean distances from each member node to the CH candidate.

$$D_j = \sum_{i \in C} d_{ij} \qquad\qquad C_j = \frac{1}{\sum_{i \in C} d_{ij}}$$

A lower value of Dj indicates that the CH is located closer to the majority of cluster members, which directly translates into reduced transmission energy for intra-cluster communication. Since radio energy consumption increases polynomial with distance, minimizing Dj is essential for achieving energy-efficient clustering and preventing uneven energy depletion among member nodes.

Complementary to this, the node centrality metric Cj is introduced to capture how centrally positioned a node is within a cluster. Centrality is typically defined as the inverse of the cumulative distance between the candidate CH and other cluster members. A higher value of Cj implies that the node is geometrically closer to most nodes in the cluster, making it a more favorable choice for the CH role. Central nodes reduce average communication distance, improve cluster compactness and enhance overall energy balance. To jointly optimize energy efficiency and spatial efficiency, recent clustering algorithms formulate CH selection as a multi-criteria optimization problem. This is achieved through a utility function that combines normalized residual energy and spatial centrality metrics. A typical utility function assigns higher scores to nodes that possess both sufficient residual energy and advantageous spatial positioning. Weighting factors are introduced to control the relative importance of energy and spatial criteria, enabling flexible adaptation to different network requirements. The utility function is expressed as:

$$\max \sum_{j=1}^{N} x_j \left(w_1 \frac{E_j}{E_{max}} + w_2 \frac{C_j}{C_{max}} \right)$$

By maximizing this utility function under communication cost constraints, the clustering algorithm selects CHs that minimize intra cluster communication overhead while ensuring balanced energy consumption across the network. Such formulations significantly improve network lifetime, reduce the formation of energy holes and provide a more uniform distribution of CHs compared to purely energy-based or random selection mechanisms. These spatially aware utility driven approaches form an important foundation for hybrid clustering frameworks and are particularly relevant for large scale and irregular environments such as underground mining WSNs.

2.3 Meta-Heuristic Optimization Based Approaches

Meta-heuristic algorithms have been extensively applied to address the NP-hard clustering and routing problems inherent in WSNs, where the objective is to optimize multiple conflicting performance metrics under strict resource constraints. The clustering problem in WSNs determining the optimal number and placement of cluster heads, cluster membership and routing paths belongs to a class of combination optimization problems that become computationally intractable as network size increases. Similarly, energy efficient routing requires optimal path selection under dynamically changing energy levels, link qualities and traffic conditions, further increasing problem complexity. To overcome these challenges, researchers have employed meta heuristic optimization techniques such as PSO, GA,
25 ACO, SA and DE. These algorithms are inspired by natural phenomena and are capable of exploring large and complex search spaces to find near optimal solutions within acceptable computational time. In the context of WSNs, meta-heuristics are typically used to optimize cluster head selection, minimize intra and inter cluster communication costs, balance energy consumption and extend overall network lifetime.

Meta-heuristic based clustering algorithms formulate CH selection as a global optimization problem, where candidate solutions represent possible CH configurations. Fitness functions are designed to incorporate multiple objectives, such as residual energy, node-to-CH distance, network coverage and load balancing. By iteratively refining candidate solutions through evolutionary or swarm based
29 operators, these algorithms achieve significant improvements in network stability and energy efficiency compared to classical heuristic protocols. Despite their effectiveness, meta-heuristic approaches often incur high computational complexity

and convergence time, particularly in large-scale or dynamic WSNs. Many such algorithms assume static network conditions and require centralized execution or offline optimization, limiting their adaptability to real time changes such as node mobility, energy depletion, or link variability. These limitations highlight the need for more adaptive and lightweight solutions, motivating the integration of learning based approaches that can provide real time decision-making with lower overhead.

2.3.1 Particle Swarm Optimization (PSO)

In PSO based clustering, the collective behavior of a swarm of particles is exploited to identify near-optimal CH locations that minimize communication energy and improve network performance. Each particle in the swarm represents a potential clustering solution, typically encoding the positions or indices of selected CHs. Particles iteratively adjust their positions in the search space by learning from their own historical best solution (personal best) and the best solution found by the swarm (global best). This cooperative learning mechanism enables PSO to efficiently explore complex and high dimensional solution spaces associated with WSN clustering problems. The fitness function in PSO based clustering is usually designed to minimize intra cluster distances, reduce transmission energy and balance the energy load among nodes. By placing CHs closer to their member nodes, PSO significantly lowers the energy consumed during intra-cluster communication, which constitutes a major portion of overall network energy expenditure. As a result, PSO based protocols often achieve improved network lifetime, enhanced stability period and better load distribution compared to classical clustering methods such as LEACH. This models the cooperative movement patterns observed in bird flocking and fish schooling [24]. Each member of the population, known as a particle, represents a possible solution. These particles move within the search space, updating their positions according to their own best performance and the best performance observed among all particles. Over iterations, particles converge toward the best solution.

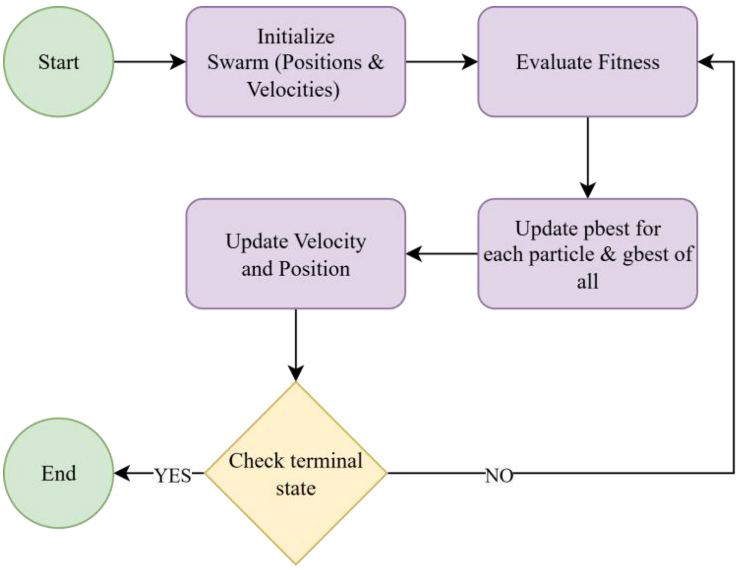


Figure 2.3.1 Flow chart of the PSO algorithm [14]

19

This figure represents the working principle PSO algorithm. Initially, a group of particles is generated, each symbolizing a candidate solution with randomly assigned positions and velocities. At each iteration, the fitness of all particles is evaluated using the objective function. Each particle then updates its personal best position (pbest) and the global best position (gbest) found by the entire swarm [30]. The velocity and position of each particle are updated according to the below equations:

$$v_i(t+1) = w \cdot v_i(t) + c_1 \cdot r_1 \cdot (pbest_i - x_i(t)) \\ + c_2 \cdot r_2 \cdot (gbest - x_i(t))$$

$$x_i(t+1) = x_i(t) + v_i(t+1)$$

52

where '*w*' denotes the inertia weight, '*c1*' and '*c2*' are acceleration coefficients and '*r1*', '*r2*' belongs to [0,1] are random numbers. These Equations together balance the inertia (momentum), cognitive (self-learning) and social (swarm cooperation) components of each particle. This process iteratively continues until the swarm converges toward an optimal or near-optimal solution based on the defined fitness function. Through iterative velocity and position updates, particles gradually converge toward clustering solutions that minimize intra-cluster distances, balance energy consumption and reduce transmission cost. While these update functions provide strong optimization capability, their reliance on repeated global updates and fitness evaluations contributes to computational complexity and communication overhead, particularly in large-scale or dynamic WSN environments. This limitation motivates the integration of more adaptive and lightweight approaches, such as reinforcement learning based hybrid clustering frameworks.

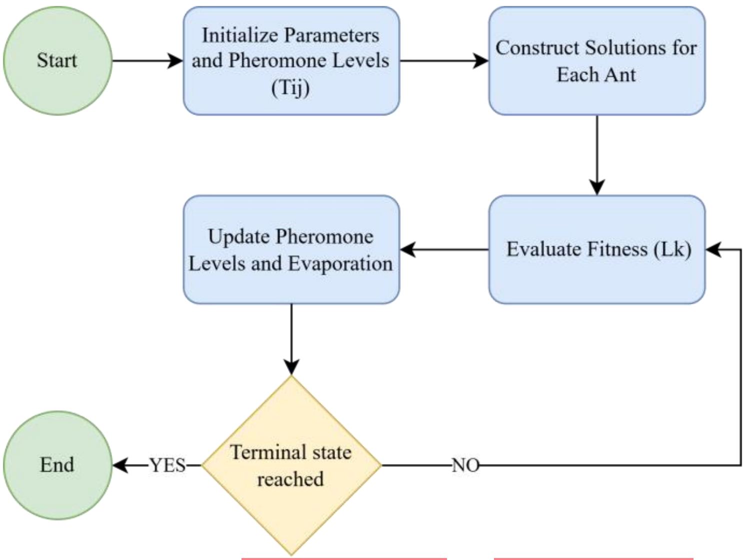
Despite these advantages, PSO based clustering algorithms generally require iterative global updates to converge toward an optimal or near optimal solution. Each iteration involves fitness evaluation and position updates for all particles, which can introduce substantial computational overhead. Moreover, PSO implementations frequently assume centralized control and global network knowledge, making them less suitable for large scale, highly dynamic, or resource constrained WSN deployments. These limitations restrict the practical applicability of PSO in real time and safety critical environments, motivating the exploration of hybrid and learning based clustering frameworks that offer adaptability with reduced overhead.

60

2.3.2 Genetic Algorithms (GA) and Ant Colony Optimization (ACO)

GA model the CH selection problem as a global optimization task, where each chromosome represents a possible clustering configuration and the fitness function typically incorporates residual energy, intra cluster distance, communication cost and load distribution. By iteratively applying selection pressure toward high fitness solutions and introducing diversity through crossover and mutation, GA can effectively mitigate energy holes and uneven CH assignments. Nevertheless, this global search paradigm requires maintaining and evaluating a large population of candidate solutions across many generations. In dense or large scale WSNs, this translates into substantial computational effort, increased memory usage and higher control overhead, making real time execution on resource constrained sensor nodes impractical. As a result, GA based CH selection is often relegated to centralized base stations, which introduces additional communication latency and a single point of failure. Similarly ACO formulates routing and clustering as a probabilistic path finding process, where artificial ants traverse the network to discover energy efficient routes based on pheromone intensity and heuristic information such as distance or residual energy. Over time, frequently used high quality paths receive stronger pheromone reinforcement, enabling emergent optimization of routing structures and cluster organization.

However, in highly dynamic environments such as mobile WSNs or underground mining scenarios with frequent link quality variations the pheromone information can quickly become outdated. This necessitates frequent re-initialization or accelerated evaporation, further increasing overhead and destabilizing convergence behavior. Another critical limitation of both GA and ACO methods lies in their sensitivity to
54 parameter tuning. Performance strongly depends on carefully selected parameters, including population size, crossover and mutation rates in GAs and pheromone evaporation rates, exploration exploitation balance and ant population size in ACO. Determining optimal parameters often requires extensive offline experimentation, reducing the adaptability of these techniques.



13 Figure 2.3.2 Flow chart of the ACO algorithm

13 This figure show the working principle of ACO algorithm. It simulates artificial ants that iteratively construct solutions using pheromone trails and heuristic information. The process begins with the initialization of parameters and pheromone levels on all paths. During solution construction, each ant builds a solution based on the probability of selecting the next path. After all ants complete their paths, the pheromone update phase reinforces paths used by better solutions while allowing pheromone evaporation to prevent premature convergence and find optimal path. The
30 algorithm proceeds in an iterative manner until a stopping criterion such as reaching the maximum number of iterations or achieving convergence is met [31]. The core equations of ACO are the path selection probability probability are expressed as :

$$P_{ij}^{k}(t) = \begin{cases} \dfrac{[\tau_{ij}(t)]^{\alpha} \cdot [\eta_{ij}]^{\beta}}{\sum\limits_{l \in N_i^k} [\tau_{il}(t)]^{\alpha} \cdot [\eta_{il}]^{\beta}}, & \text{if } j \in N_i^k \\ 0, & \text{otherwise} \end{cases}$$

33 where *Pij* is probability that ant *k* moves from node *i* to node *j*, *Tij* is pheromone
46 concentration on edge (i,j) at time *t*. *ηij* is equals to *1/dij* that is heuristic information (inverse of distance or cost). α, β are control parameters for pheromone and heuristic influence.

$$\tau_{ij}(t+1) = (1-\rho) \cdot \tau_{ij}(t) + \sum_{k=1}^{m} \Delta\tau_{ij}^{k}(t)$$

23 This equation represents pheromone update rule, where *ρ* represents pheromone evaporation rate (0<ρ<1), m is total number of ants and *ΔTij* amount of pheromone deposited by ant k on edge (i,j)

$$\Delta\tau_{ij}^{k}(t) = \begin{cases} \dfrac{Q}{L_k}, & \text{if ant } k \text{ uses edge } (i,j) \\ 0, & \text{otherwise} \end{cases}$$

This represents pheromone deposition amount, where *Q* is pheromone constant and *Lk* is total cost or path length of the tour constructed by ant *k*. Furthermore, GA and ACO based approaches generally lack inherent mechanisms for continuous online learning. Once an optimal or near-optimal solution is obtained, adapting to node failures, energy depletion, or environmental changes typically requires rerunning the optimization process from scratch or with limited reuse of prior solutions. This restart-based adaptation is inefficient for long-lived networks and conflicts with the real-time decision-making requirements of safety-critical applications. Although evolutionary and swarm intelligence techniques such as GA and ACO offer strong global optimization capabilities for CH selection and routing, their computational intensity, slow convergence in dynamic conditions, reliance on centralized or offline processing and limited online adaptability significantly constrain their practical applicability.

2.4 Fuzzy Logic and Multi Criteria Clustering

Fuzzy logic based clustering approaches address the inherent uncertainty, imprecision and nonlinear behavior of WSN by enabling multi criteria decision making for CH selection. Instead of relying on strict threshold based or probabilistic models, fuzzy inference systems (FIS) utilize linguistic variables and membership functions to represent network parameters such as residual energy, node density, distance to the cluster centroid or base station, link quality and traffic load. By combining these inputs through a set of fuzzy rules, protocols such as MRCH and various fuzzy LEACH derivatives can make more balanced and context aware CH selection decisions, thereby reducing frequent CH re elections and uneven energy depletion. The integration of fuzzy logic allows these protocols to dynamically adapt to heterogeneous network conditions and gradual changes in node energy or topology.

As a result, fuzzy based clustering schemes have demonstrated improvements in key performance metrics, including pdr, network stability period, throughput and overall lifetime. Their ability to smoothly trade off conflicting objectives such as minimizing communication distance while maximizing residual energy makes them particularly effective in environments where deterministic models fail to capture real world variability. Despite these advantages, fuzzy logic based clustering methods suffer from notable practical limitations. The design of an effective fuzzy inference system requires a carefully crafted rule base and appropriately tuned membership functions, which often depend on expert knowledge and extensive offline experimentation. As the number of input parameters increases, the rule base grows exponentially, leading to higher computational complexity and memory requirements. This rule explosion problem becomes especially critical in large-scale or heterogeneous WSNs where diverse node conditions must be evaluated.

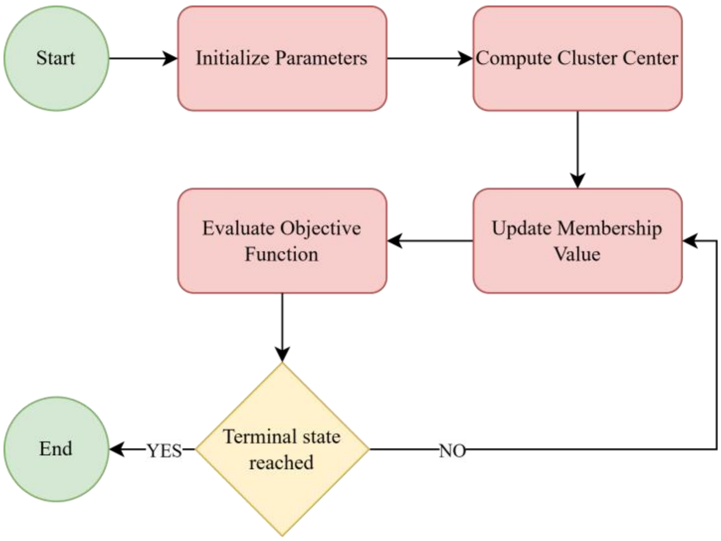


Figure 2.4 Flow chart of the FCM algorithm

This figure illustrates the working principle of the FCM algorithm. Here the process begin with the initialization of the membership matrix U. It assigns random membership values to each data point for all clusters. Next, cluster centers are computed based on the current membership values. Then the algorithm updates the membership degrees for each data point using the updated cluster centers. Later on the objective function is evaluated to measure clustering performance. If the change in membership values or cluster centers between iterations is smaller than a predefined threshold, the process is said to have converged, otherwise, the algorithm repeats the update steps. Once convergence is achieved, the final cluster assignments are generated, representing the optimal fuzzy partition of the dataset

$$J_m = \sum_{i=1}^{N} \sum_{j=1}^{C} u_{ij}^m \, \|x_i - c_j\|^2 \qquad\qquad c_j = \frac{\sum_{i=1}^{N} u_{ij}^m \, x_i}{\sum_{i=1}^{N} u_{ij}^m}$$

These equation represents the objective function that FCM seeks to minimize. It measures the total weighted distance between all data points *xi* and the cluster centers *cj*. The weight is given by the membership value *Uij,* m which indicates how strongly a data point belongs to a particular cluster. It is used to update the cluster center. Each cluster center is calculated as the weighted mean of all data points, where the weights are the membership degrees raised to the power *m*.

$$u_{ij} = \frac{1}{\sum_{k=1}^{C} \left(\frac{\|x_i - c_j\|}{\|x_i - c_k\|} \right)^{\frac{2}{m-1}}}$$

This equation defines how the membership value *Uij* of each data point to each cluster is updated. From equation we can conclude that the membership is inversely related to the distance between the data point and the cluster center. The ratio term ensures that all membership values for a data point sum to 1 across all clusters for maintaining normalization

Moreover, executing fuzzy inference comprising fuzzification, rule evaluation, aggregation and defuzzification introduces non-negligible processing overhead and energy consumption. For ultra-low-power sensor nodes with limited processing capability, memory and battery capacity, this overhead can offset the energy savings gained from improved clustering decisions. Consequently, many fuzzy based approaches rely on centralized or cluster level processing, which increases communication overhead and reduces fault tolerance. In addition, most fuzzy logic based clustering protocols lack autonomous learning or self optimization capabilities. Once the rule base and membership functions are defined, they remain static throughout network operation, limiting adaptability to long term environmental changes, evolving traffic patterns, or progressive energy heterogeneity. Updating or re-configuring the fuzzy system typically requires manual intervention or offline retraining, which is impractical for long lived or safety critical deployments.

2.5 Reinforcement Learning Based Approaches in WSNs

RL has gained significant attention as a promising solution for adaptive and intelligent WSN management. It is based on the concept of interaction between an intelligent agent and its surrounding environment. In WSN applications, sensor nodes, cluster heads or centralized controllers can be modeled as learning agents that observe the current network state, perform actions such as CH selection, routing decisions or transmission power adjustment and receive feedback in the form of rewards or penalties. Through continuous interaction with the network environment, the agent learns an optimal policy that maximizes long term performance objectives such as network lifetime, energy efficiency, throughput and communication reliability [17].

Several RL algorithms have been applied in WSN research. Among them, Q-Learning is one of the most widely adopted model free RL techniques. In Q-Learning based clustering protocols, each node maintains a Q-table that stores the expected reward for selecting specific actions under different network states. Over multiple communication rounds, nodes update their Q-values based on observed rewards, gradually learning energy efficient cluster head selection and routing strategies. Q-Learning based protocols have demonstrated improvements in network lifetime, stability period and throughput compared to classical clustering methods.

2.5.1 Q-Learning and Policy-Based Methods

The policy based RL methods and deep reinforcement learning techniques have also been explored for WSN optimization. These approaches utilize neural networks or policy gradient methods to approximate optimal decision policies, enabling efficient operation in large scale networks with complex state spaces. Deep Q-Networks (DQN) and Actor-Critic frameworks have shown promising results in adaptive routing, topology control and energy management. Algorithms such as Q-learning, SARSA and DQN have been employed for CH selection, routing and topology control [17]. Protocols like EER-RL dynamically adjust CH roles based on learned energy patterns, improving adaptability to changing network conditions and delivers significant improvements. However, such approaches typically require higher computational and memory resources, making them challenging to deploy on low power sensor nodes. Q-Learning is represented by following equation.

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

Where Q(s,a) denotes the action value function, which predicts the expected total reward an agent can achieve by taking action a in state s and then following the optimal policy afterward. The learning rate α, which ranges between 0 and 1, controls how much the newly gained information influences the existing value estimates, balancing the trade off between learning speed and stability. The immediate reward received after performing action 'a' is represented by r, while the discount factor γ(gamma), also between 0 and 1, determines the relative importance of future rewards as compared to immediate ones [18]. The term Q(s′,a′) represents the maximum estimated value of the next state s′ over all possible actions a′, which helps the agent make decisions that maximize long term returns. The difference between the target value [$r+\gamma \max Q(s',a')$] and the current estimate $Q(s,a)$ is referred to as the Temporal Difference (TD) error [18] which quantifies the discrepancy between predicted and actual outcomes. By iteratively applying this update after every round of operation, each node refines its action-value estimates, allowing it to improve CH selection decisions over time.

2.5.2 Limitations of Centralized RL

In centralized reinforcement learning frameworks, a single controller, typically located at the base station or a central processing node, collects global network information and performs policy learning for cluster head selection, routing optimization and resource allocation. While this approach simplifies algorithm implementation and allows access to complete network state information, it introduces several critical limitations that restrict its practical applicability in real-world WSN deployments, particularly in harsh and dynamic environments such as underground mining [18][19].

One of the major limitations of centralized RL approaches is scalability. As the network size increases, the amount of state information required by the centralized controller grows significantly. Large-scale WSNs may consist of hundreds or thousands of sensor nodes, each generating information related to residual energy, communication links, traffic load and spatial positioning [20]. Aggregating and processing this vast amount of information at a single controller results in excessive computational complexity and increased decision latency. Consequently, centralized RL models become inefficient and difficult to scale for dense or geographically distributed sensor deployments.

Another significant challenge is communication overhead. Centralized RL requires frequent transmission of state information from sensor nodes to the central controller and dissemination of control decisions back to the network. This bidirectional communication generates substantial control packet overhead, which consumes additional energy and reduces the overall network lifetime. In energy-constrained WSN environments, excessive communication overhead may negate the energy savings achieved through intelligent clustering and routing optimization [19] [20].

Centralized RL frameworks also suffer from single point of failure vulnerability. Since learning and decision-making are dependent on a central controller, failure of the base station or communication link disruption can lead to complete network performance degradation [21]. This limitation is particularly critical in safety-sensitive applications such as underground mining monitoring, where continuous network operation is essential for hazard detection and worker safety.

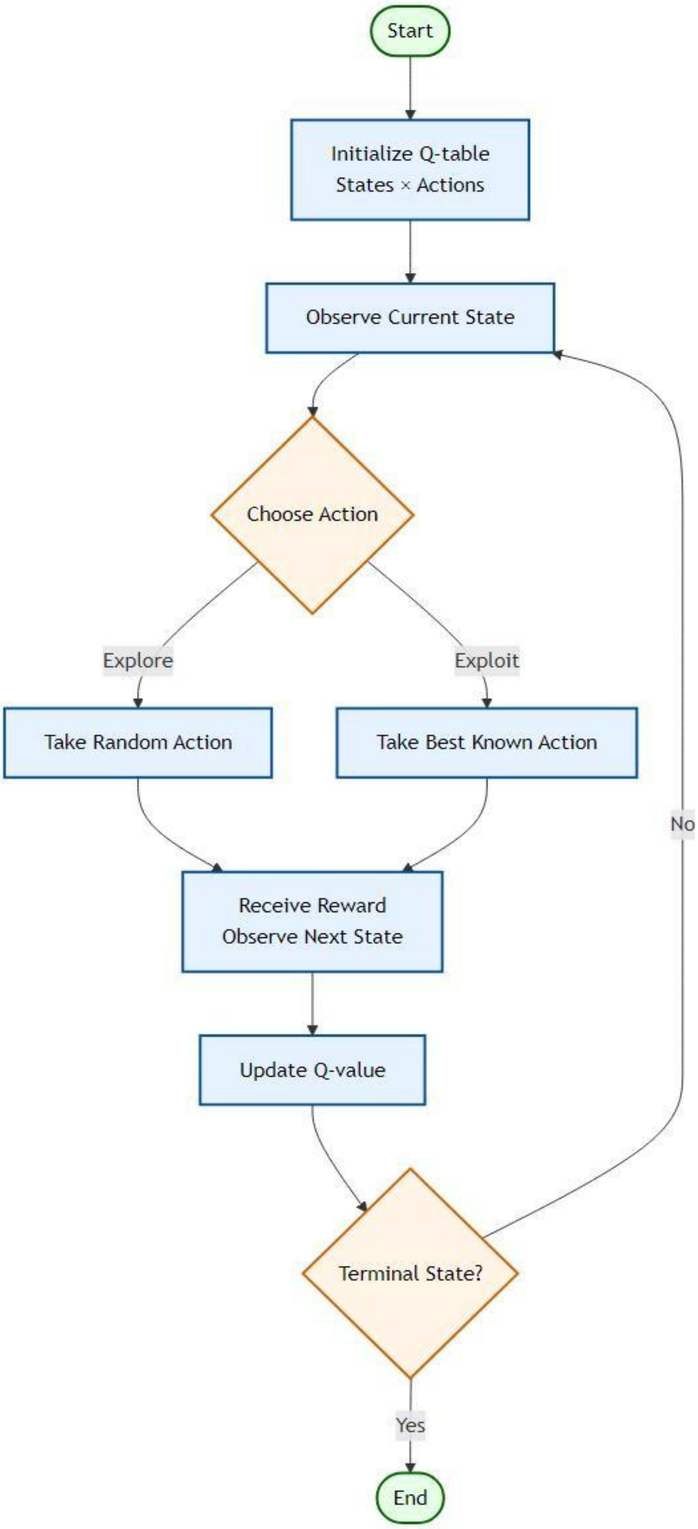


Figure 2.5 Flow chart of the Q-Learning algorithm

2.6 Hybrid Clustering and Intelligent Frameworks

Hybrid approaches aim to combine the strengths of classical clustering protocols, meta heuristic optimization techniques, fuzzy logic and RL to enhance the performance of WSN. Multi-layer hybrid frameworks typically integrate energy-aware cluster head selection, spatial clustering techniques such as K-Means or fuzzy clustering and RL based adaptive decision making mechanisms. These integrated approaches enable simultaneous optimization of multiple performance metrics. Protocols such as Energy Optimized Cluster and Grid Structure (EOCGS) determine the optimal number of clusters and grid heads to achieve balanced energy utilization across the network [19]. Recent studies have demonstrated that hybrid intelligent frameworks significantly outperform conventional standalone clustering methods by providing improved adaptability, enhanced energy efficiency and prolonged network stability under dynamic network conditions.

2.7 Critical Analysis and Research Gaps

Despite substantial progress in energy efficient clustering and routing techniques for WSN, existing approaches continue to face several critical challenges. Meta-heuristic optimization methods provide strong global search capability but incur high computational complexity and convergence delays, making them unsuitable for real-time deployment in resource constrained sensor networks. Classical clustering protocols often suffer from uneven energy consumption and early node depletion due to probabilistic or static cluster head selection strategies. RL based models introduce adaptive decision making capabilities, however, many existing implementations rely heavily on centralized learning architectures, which suffer from scalability limitations, communication overhead and vulnerability to single point failures.

Existing challenges and RLHC Approach

- Early node depletion in classical clustering protocols: RLHC applies energy-aware cluster head selection to ensure balanced energy consumption.

- Scalability and communication overhead in centralized RL models: RLHC adopts a distributed adaptive learning mechanism to improve scalability and robustness.

- Static decision-making in existing Q-learning approaches: RLHC introduces dynamic state monitoring including residual energy, cluster load and communication reliability for adaptive optimization.

- Lack of spatial awareness in traditional clustering techniques: RLHC integrates EEKA and K-Means clustering to ensure reduced communication distance.

- Poor adaptability to underground mining environments: RLHC provides continuous learning based adaptation to handle dynamic and harsh underground network conditions.

2.8 Summary of Energy-Efficient WSN Studies

Table 2.1 summarizes key energy efficient clustering and routing studies, highlighting their techniques and identified research gaps.

SL No	Paper & Year	Optimization Technique Used	N/w Lifetime Improved	Limitations / Research gap
1	Zhang et al. (2021) Energy-Efficient Clustering Routing Protocol Based on Weight for WSN in Coal Mine[5]	Weighted Clustering (Residual energy and Distance factors)	Improved by balancing energy in cluster heads	Does not adapt for dynamic topological
2	Li et al. (2021) WSN Coverage Optimization in Coal Mine Based on Improved Whale Optimization Algorithm[6]	Whale Optimization Algorithm (WOA)	Extended by minimizing redundant coverage	High computational complexity for low-power sensor nodes
3	Wang & Liu (2021) A Hybrid Energy-Efficient Routing Protocol for Underground Coal Mine Monitoring[7]	Hybrid Routing (Combining Tree-based and Cluster-based)	Prolonged network life in large scale linear topologies	Complexity in switching between modes leads to latency
4	Chen et al. (2022) Research on Energy Balanced Routing Strategy for Coal Mine WSN based on Mobile Sink[8]	Mobile Sink (Sink moves to collect data)	Improves lifetime by 20% vs LEACH.	Requires precise path planning for the mobile vehicle.
5	Kumar & Pal (2022) Fuzzy Logic Based Clustering Algorithm for WSN in Underground Mines[9]	Fuzzy Logic Controlle	Better decision making for CH selection	Rule-base design is complex and static.
6	Al-Kaseem et al. (2022) Optimized Energy-Efficient Path Planning with Multiple Mobile Sinks[10]	Ant Colony Optimization + Genetic Algorithm	66% vs. standard LEACH	Heavy computation demand Static environment
7	Raed et al. (2022) Efficient Energy Mechanism for Underground Mining Monitoring[11]	Enhanced LEACH + DEECP	25% vs standard LEACH	No mobility support Static environment
8	Salman et al. (2022) Optimization of LEACH Protocol for WSNs in Terms of Energy Efficiency and Network Lifetime	Particle Swarm Optimization	30% vs. standard LEACH	No mobility support Non heterogeneous environmnt

SL No	Paper & Year	Optimization Technique Used	N/w Lifetime Improved	Limitations / Research gap
9	Gamal et al. (2022) Enhancing Lifetime of WSNs Using Fuzzy Logic LEACH and Particle Swarm Optimization[13]	Fuzzy + Particle Swarm Optimization	18% vs. standard LEACH	Increased control overhead from fuzzy inference
10	Mohapatra et al. (2022) Mobility Induced Multi-Hop LEACH Protocol in Heterogeneous Mobile Network[14]	Modified multi-hop LEACH	20% vs. standard LEACH	Assumes uniform mobility Non heterogeneous environmnt
11	Liu et al. (2023) Improved Grey Wolf Optimizer (GWO) for 3D Node Coverage in Mines[16]	Grey Wolf Optimizer (GWO)	Maximize s coverage in 3D laneways	Does not fully account for signal multipath fading in tunnels
12	Adnan et al. (2023) LoRa-based Wireless Underground Sensor Networks: Evaluation and Deployment[17]	LoRa (Long Range) Modulation (Physical layer optimization)	Extend battery life due to low power usage.	Low data rate High Packet collision probability.
13	Wu et al. (2023) Fault Tolerant Routing Mechanism for Underground Coal Mine WSN[18]	Multipath Routing (Redundant paths)	Increases reliability;	Higher energy cost due to redundant data transmission
14	Yang et al. (2023) Dynamic Duty Cycling Scheme for Underground Mine Disaster Monitoring[19]	Dynamic Duty Cycling (Sleep/Wake scheduling)	Drastically reduces idle listening energy usage.	Increased delay in event detection and synchronization issues in deep tunnels
15	Bibi et al. (2023) Energy Harvesting WSN Routing Protocol for Underground Applications[20]	Energy Harvesting Aware Routing	Extend lifetime if harvesting matches usage	Harvesting sources are intermittent requires battery
16	Tadele A. Abose et al. (2023) Optimized Cluster Routing Protocol With Energy-Sustainable Mechanisms for Wireless Sensor Networks[21]	Stable Election Protocol	36% vs standard LEACH	No Multi-hopping Homogenous network
17	Aleem & Thumma (2025) Hybrid DEEC EEKA RL in IoT- WSNs[23]	Residual energy ratio (DEEC) EEKA constraints + RL	17.5% vs standard LEACH	Computational overhead, No mobility support

Chapter 3

Problem Statement

WSN have become a fundamental technology for real time monitoring and automation in hazardous and inaccessible environments such as underground mining, industrial monitoring and disaster management. In underground mining environments, sensor nodes are deployed to monitor critical parameters including gas concentration, temperature, humidity and structural stability. Reliable and continuous communication among sensor nodes is essential to ensure early hazard detection and worker safety. However, the deployment of WSNs in underground environments introduces several technical challenges related to energy efficiency, communication reliability, network scalability and adaptability to dynamic environmental conditions.

Additionally underground mining WSN deployments present deployment challenges due to irregular tunnel geometries, severe signal attenuation, fluctuating channel conditions and dynamic node mobility caused by mining operations. These factors lead to unpredictable energy depletion patterns, communication instability and uneven cluster formation. Existing clustering and optimization techniques are not specifically designed to handle such complex and safety-critical deployment conditions.

Also sensor nodes in WSNs are typically powered by limited battery resources and replacing or recharging these batteries in underground mining environments is often impractical and costly. Consequently, efficient energy utilization becomes the most critical design constraint for ensuring prolonged network operation. Traditional clustering protocols such as LEACH and DEEC attempt to improve energy efficiency by organizing sensor nodes into clusters and rotating cluster head responsibilities. Although these protocols reduce communication overhead, they rely on probabilistic or static cluster head selection strategies that often result in uneven energy consumption and premature node failure.

Therefore, there is a need for an adaptive, multi objective and energy efficient clustering framework that can dynamically optimize cluster formation, communication reliability and network lifetime while maintaining computational efficiency suitable for resource constrained sensor nodes.

The proposed Reinforcement Learning with Hybrid Clustering (RLHC) framework aims to address these challenges by introducing a multi-layer clustering architecture capable of dynamically selecting cluster heads, optimizing cluster distribution and adapting network communication policies based on real time environmental feedback. The objective is to improve energy efficiency, extend network lifetime, enhance packet delivery reliability and provide scalable and robust WSN operation in harsh underground mining environments.

Chapter 4

Proposed Solution

The proposed Reinforcement Learning with Hybrid Clustering (RLHC) framework introduces a hierarchical multi-layer architecture designed to address energy imbalance, inefficient cluster formation and unreliable communication in underground wireless sensor network deployments.

As illustrated in Fig. 4.1 End-to-End Layer Integration, RLHC integrates deterministic optimization methods with adaptive learning mechanisms. Each layer progressively refines network organization, beginning from energy-aware candidate selection to intelligent communication management.

The RLHC framework consists of five sequential layers:

i. Distributed Energy Efficient Clustering Protocol (DEEC)
ii. Energy Efficient Knapsack Algorithm (EEKA)
iii. K-Means Spatial Clustering
iv. Reinforcement Learning using Q-learning
v. Communication Management Layer

Each layer performs a specific optimization task while passing refined information to the subsequent stage.

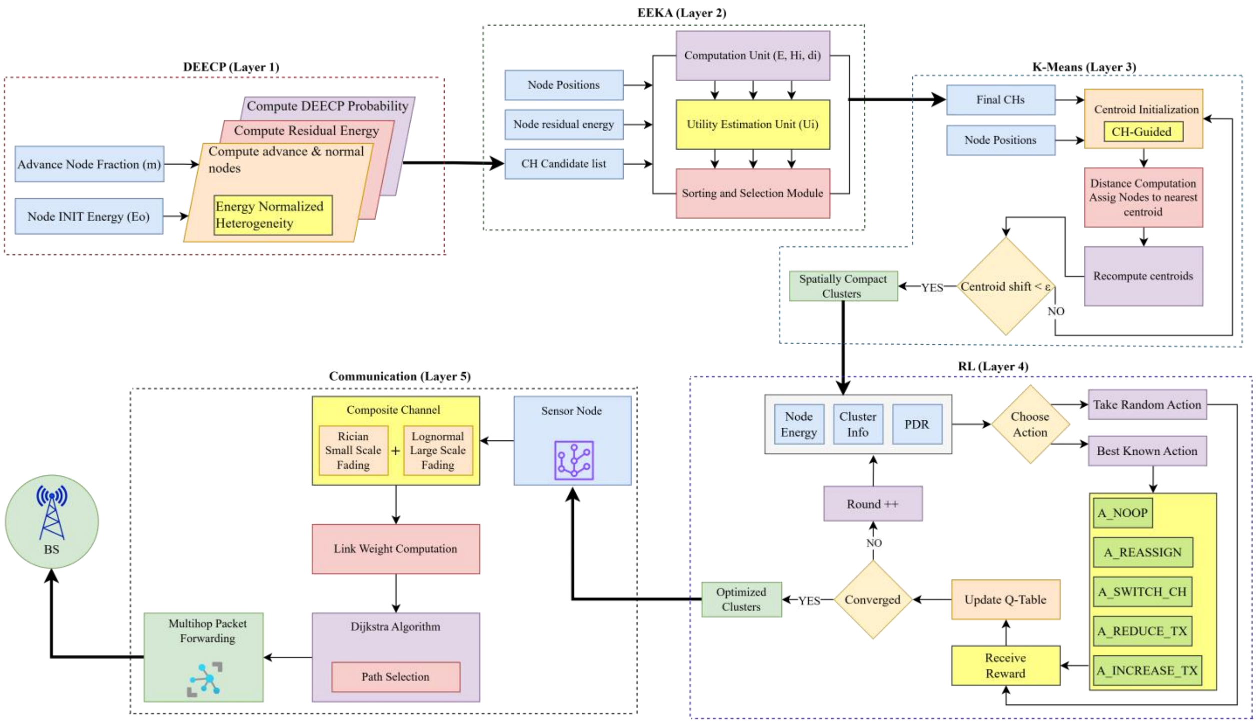


Figure 4.1 End-to-End Layer Integration

4.1 Distributed Energy Efficient Clustering Protocol (DEEC)

The first layer of RLHC generates an energy-aware pool of candidate cluster heads using the Distributed Energy Efficient Clustering Protocol (DEEC).
Wireless sensor networks deployed in underground mining environments exhibit heterogeneous energy distributions due to varying node roles and sensing workloads. Traditional probabilistic clustering schemes fail to account for residual energy variations, resulting in premature node failures.
DEEC addresses this limitation by assigning cluster head selection probability proportional to the residual energy of each node [22].

The probability of node i becoming a cluster head is defined as Pi:

$$P_i = prop \times \left(\frac{E_i}{\bar{E}(r)} \right) \qquad \bar{E}(r) = \frac{E_{\text{total}} \cdot \left(1 - \frac{r}{R}\right)}{N}$$

$$T(i) = \begin{cases} \frac{P_i}{1 - P_i \cdot \left(r \bmod \left(\frac{1}{P_i}\right)\right)}, & \text{if } i \in G \\ 0, & \text{otherwise} \end{cases}$$

Where Popt denotes optimal cluster head probability. Ei(r) represents residual energy. Eavg(r) denotes network average energy at round r.

Nodes possessing higher residual energy obtain increased probability of selection. The output of this layer is an energy-qualified candidate cluster head set.

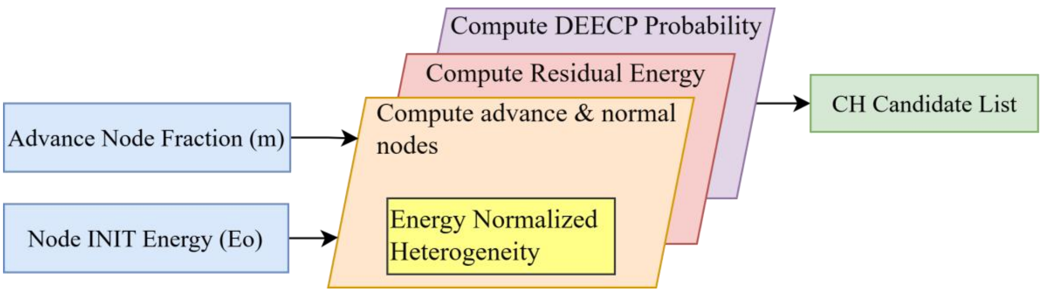


Figure 4.1.1 DEEC Protocol

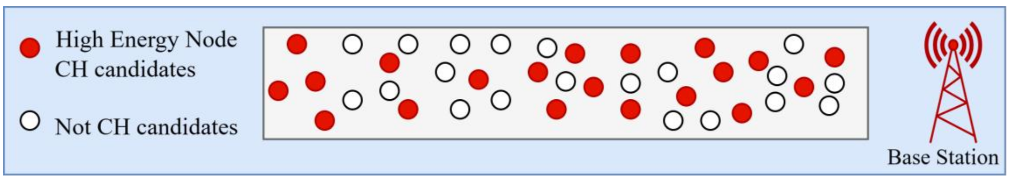


Figure 4.1.2 Candidates cluster heads

4.2 Energy Efficient Knapsack Algorithm (EEKA)

Although DEEC produces energy-efficient candidates, cluster heads may become spatially concentrated, leading to uneven coverage and communication congestion. To address this issue, RLHC employs EEKA, which selects an optimal subset of candidate CH that maximize network utility while satisfying spatial distribution constraints [23].

The optimization objective is expressed as:

$$U_i = \alpha \left(\mathrm{Norm}\left(\frac{E_i}{\bar{E}} \right) \right)^2 + \beta \,\mathrm{Norm}\left(\frac{1}{\bar{d}_i + \varepsilon} \right) - \gamma \,\mathrm{Norm}(H_i)$$

$$\begin{aligned} U_i &: \text{ Utility (fitness) score of node } i, \\ N &: \text{ Total number of nodes,} \\ \alpha, \beta, \gamma &: \text{ Weight for energy, centrality and history penalty,} \\ E_i &: \text{ Residual energy of node } i, \\ \bar{E} &= \text{ Mean total energy,} \\ \bar{d}_i &= \text{ Average distance to } k\text{-nearest neighbors,} \\ H_i &: \text{ CH history counter,} \end{aligned}$$

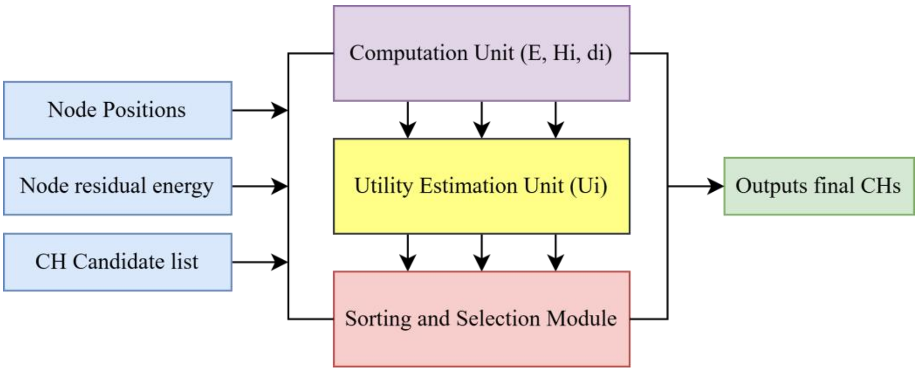


Figure 4.2.1 EEK Algorithm

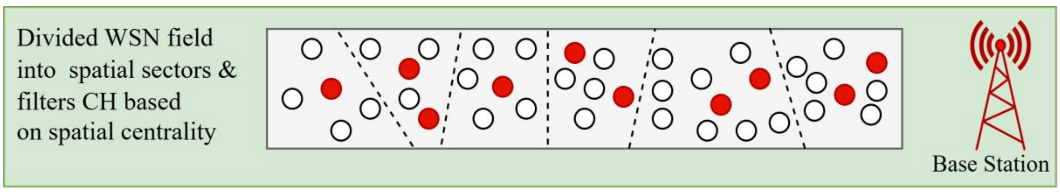


Figure 4.2.2 Spatially distributed cluster heads

1

4.3 K-Means Spatial Clustering

Following cluster head optimization, RLHC performs spatial clustering using the K-Means algorithm. The objective is to create geographically compact clusters, minimizing intra cluster communication distance and transmission energy consumption [24].
It is an iterative algorithm expressed as:

$$\mu_j^{(0)} = \mathbf{x}_{h_j} \quad \Rightarrow \quad \min_{\{\mathcal{S}_j\}_{j=1}^{k}} \sum_{j=1}^{k} \sum_{i \in \mathcal{S}_j} \left\| \mathbf{x}_i - \mu_j \right\|^2$$

$$\begin{aligned}
j &: \ j \in \{1, 2, \ldots, k\} \text{ Denotes the cluster index,} \\
k &: \ \text{Total number of clusters,} \\
\mathbf{x}_i &: \ \text{Represents the } (x_i, y_i) \text{ spatial pos. of node,} \\
\mu_j &: \ \text{Updated centroid of cluster } j, \\
\mathcal{S}_j &: \ \text{Set of sensor nodes assigned to cluster,} \\
\mathbf{x}_{h_j} &: \ \text{Spatial position of the cluster head}
\end{aligned}$$

The compact clusters reduce transmission distances and simplify learning complexity by allowing reinforcement learning decisions at cluster level instead of node level.

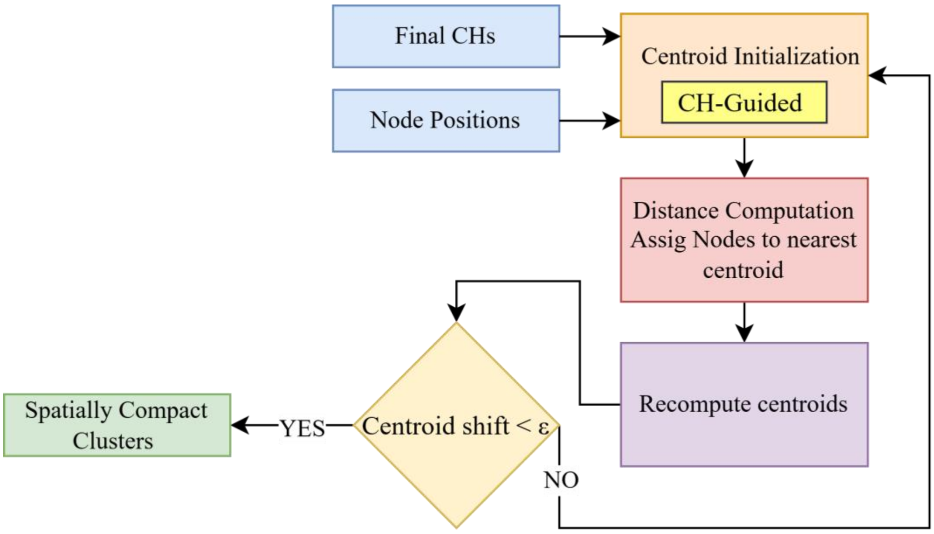


Figure 4.3.1 K-Means Algorithm

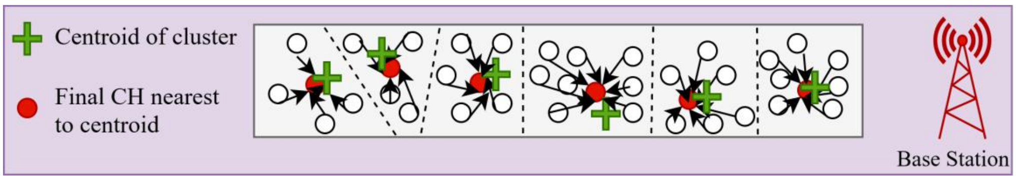


Figure 4.3.2 Spatially compact cluster

4.4 Reinforcement Learning Using Q-Learning

The fourth layer introduces intelligence into RLHC through Q-learning based adaptive optimization. Static clustering approaches cannot adapt to dynamic underground conditions such as link fading, node mobility, or traffic variations [25]. The RL agent observes network state at each communication round. Adaptive decision making by observing network metrics (energy, cluster load, PDR) and chooses adaptive actions.It learn from feedback by evaluating the outcome of each action using a reward function and updates the Q-table.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right]$$

$$Q^{\pi}(s, a) = \sum_{s'} P(s' \mid s, a) \left[R(s, a, s') + \gamma \sum_{a'} \pi(a' \mid s') Q^{\pi}(s', a') \right]$$

where α is the learning rate that determines how much new information overrides old knowledge, γ is the discount factor that defines the importance of future rewards and 'a' represents the action taken by the agent. This update allows the agent to iteratively improve its policy by balancing immediate rewards with long-term benefits, enabling more optimal decision making over time.

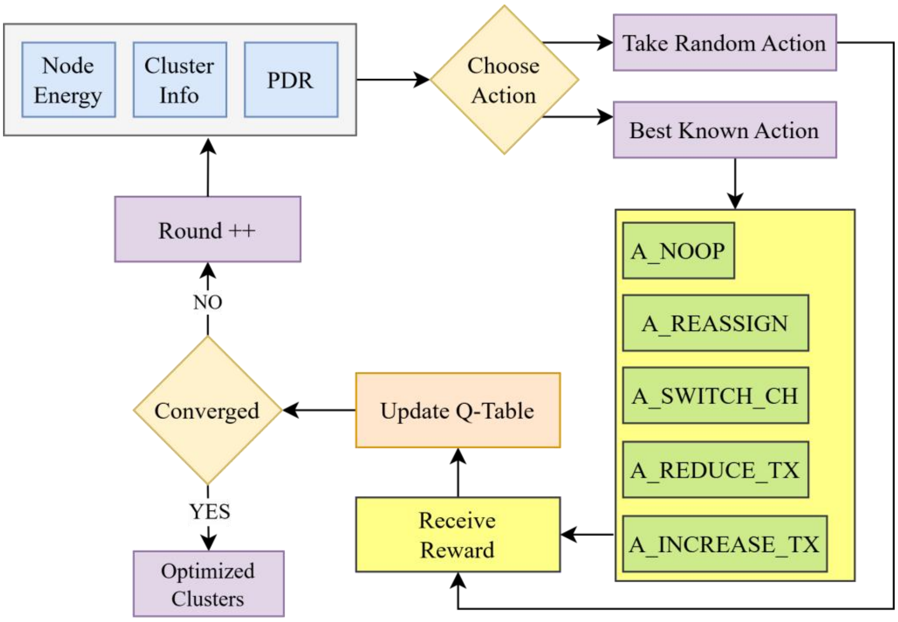


Figure 4.4.1 Reinforcement Learning Algorithm

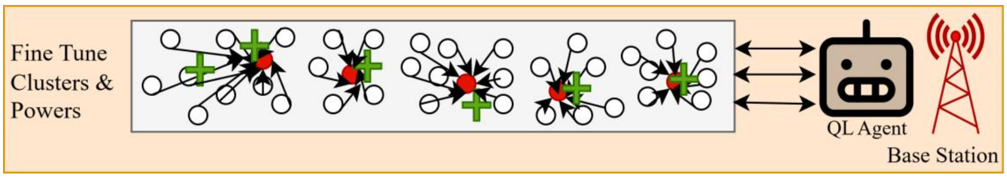


Figure 4.4.2 Centralized Q-Learning Agent Optimizing Cluster Formation

4.5 Communication Management Layer

The final RLHC layer ensures reliable data communication between sensor nodes and base station. Multi Hop routing, Data forwarding paths are determined using Dijkstra's shortest path algorithm [26]. The cost metric incorporates both distance and link reliability. Also composite channel model underground mining environments experience severe multi-path fading and shadowing. RLHC models channel behavior using Rician fading for tunnel wave guide propagation and Log-normal shadowing for obstruction losses. Received power is modeled as:

$$f_{G_C}(g) = \int_0^\infty \frac{1}{x} f_{G_R}\left(\frac{g}{x}\right) f_{G_L}(x)\, dx$$

$$G_{\text{Rician}}(d) = \sqrt{\frac{K}{K+1}} + \sqrt{\frac{1}{K+1}}X$$

$$G_{\text{LogNormal}}(d) = 10^{\frac{X_\sigma}{10}}, \quad X_\sigma \sim \mathcal{N}(0, \sigma^2)$$

$$G(d) = G_{\text{Rician}}(d) \times G_{\text{LogNormal}}(d)$$

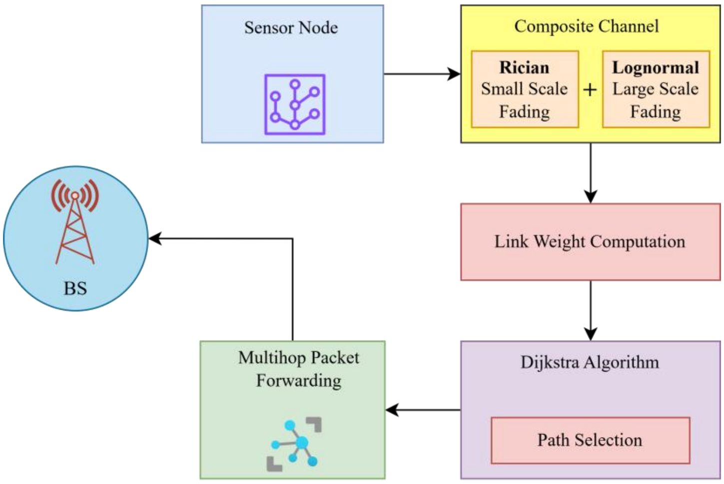


Figure 4.5.1 Communication Layer

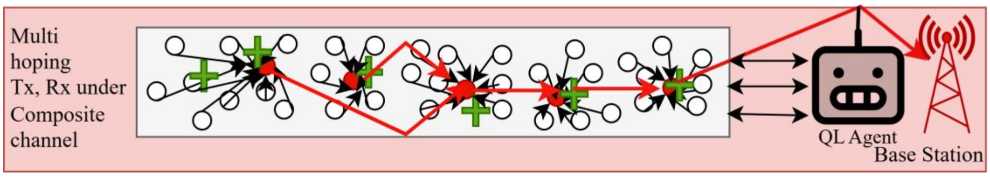


Figure 4.5.2 Multihopping communication under composite channel

The proposed solution follows an end-to-end integrated pipeline modeling approach to ensure seamless coordination between sensing, decision making, learning and communication processes. The objective of this pipeline is to jointly optimize energy efficiency, network lifetime and CH selection through reinforcement learning–based adaptation. The pipeline begins with network initialization, where sensor nodes are randomly deployed within the monitoring region and categorized according to heterogeneous energy levels such as normal, high-energy and super nodes. Each node is assigned an initial energy budget, spatial coordinates and communication parameters required for transmission and reception operations. In the state construction layer, relevant network parameters are extracted to represent the environment state for learning. These parameters typically include residual node energy, distance to the base station, neighborhood density and historical participation in cluster head selection. The state vector provides a compact representation of node health and network topology at a given round.

16

The next stage corresponds to the Q-learning decision module, where each eligible node evaluates possible actions using the learned Q-table. Actions generally correspond to becoming a cluster head or remaining a member node. The agent selects actions using an exploration–exploitation strategy such as ε-greedy selection to balance learning and performance stability. Following action selection, the cluster formation layer organizes nodes into clusters based on proximity and communication cost. Selected cluster heads broadcast advertisement messages and member nodes associate with the most energy-efficient cluster head. This step reduces redundant communication and enables localized data aggregation.

27

The data aggregation and transmission stage performs intra-cluster sensing data collection. Cluster heads aggregate received data to minimize packet redundancy before transmitting compressed information to the base station. Energy consumption is calculated using the adopted radio energy model during transmission and reception events.

16

Subsequently, the reward evaluation layer computes reinforcement signals based on performance indicators such as residual energy balance, successful packet delivery, cluster stability and network lifetime improvement. The computed reward updates the Q-values using the learning rule, enabling adaptive optimization across successive rounds.

The pipeline concludes with the network update and iteration phase, where node energies are reduced according to communication costs, dead nodes are identified and the environment transitions to the next state. This iterative learning process continues until network termination conditions are reached, such as complete node depletion or predefined simulation rounds.

15

By integrating sensing, clustering, reinforcement learning and communication management into a unified workflow, the proposed pipeline enables adaptive cluster head selection and sustained energy balancing, thereby improving overall system scalability and operational lifetime in heterogeneous wireless sensor network deployments.

4.6 RLHC Deployment - Centralized Reinforcement Learning Model

In the centralized reinforcement learning configuration, a single learning agent located at the base station or sink node maintains a global Q-table. The centralized agent collects network statistics from all clusters and performs decision making using complete system knowledge.

The advantages of centralized learning include global visibility of network topology and energy distribution. Faster convergence due to consistent learning updates. Simplified coordination among clusters.

However, centralized RL introduces communication overhead because nodes must frequently transmit state information to the sink. Additionally, the sink becomes a computational bottleneck and a potential single point of failure in large-scale deployments. Within the proposed pipeline, centralized Q-learning optimizes cluster formation by evaluating global reward trends and issuing updated policies for cluster head rotation and routing strategies. This configuration achieves strong short-term throughput performance due to globally optimized decisions.

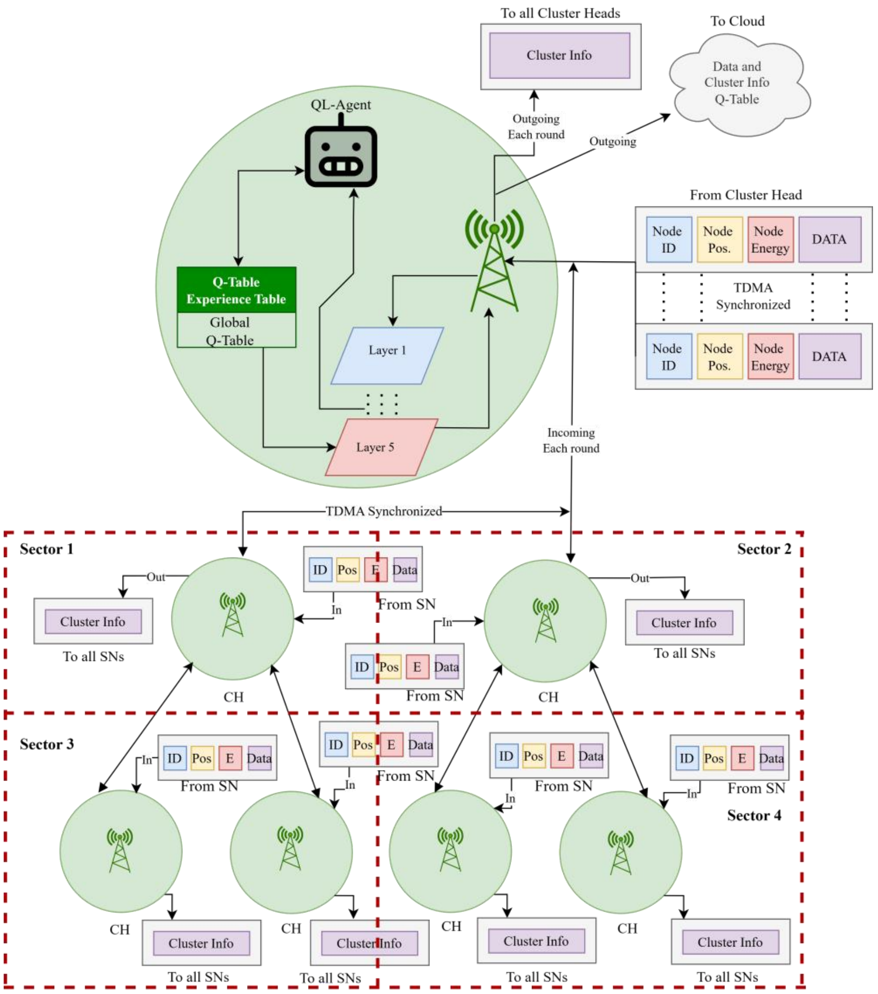


Figure 4.6.1 Centralized RLHC Deployment

4.7 RLHC Deployment - Federated Reinforcement Learning Model

To address scalability and communication overhead limitations, the proposed framework incorporates a federated reinforcement learning mechanism. In the federated configuration, multiple local learning agents operate independently within sectors or clusters. Each sector trains its own Q-table using locally observed states such as energy depletion rate, link quality and traffic load. Instead of transmitting raw sensing data or full network states, only learned model parameters or Q-value updates are periodically shared with a coordinating entity for aggregation.

The aggregated global model is redistributed back to participating agents, enabling collaborative learning while preserving decentralized autonomy. The federated approach provides several advantages, reduced communication overhead due to local learning, improved scalability for large-area deployments, enhanced fault tolerance since learning continues even if some sectors fail and better adaptability to localized environmental variations.

Simulation outcomes demonstrate that federated RLHC achieves improved sector survivability and longer network lifetime compared to the centralized variant. The distributed learning structure maintains balanced energy consumption across sectors while sustaining reliable packet delivery performance.

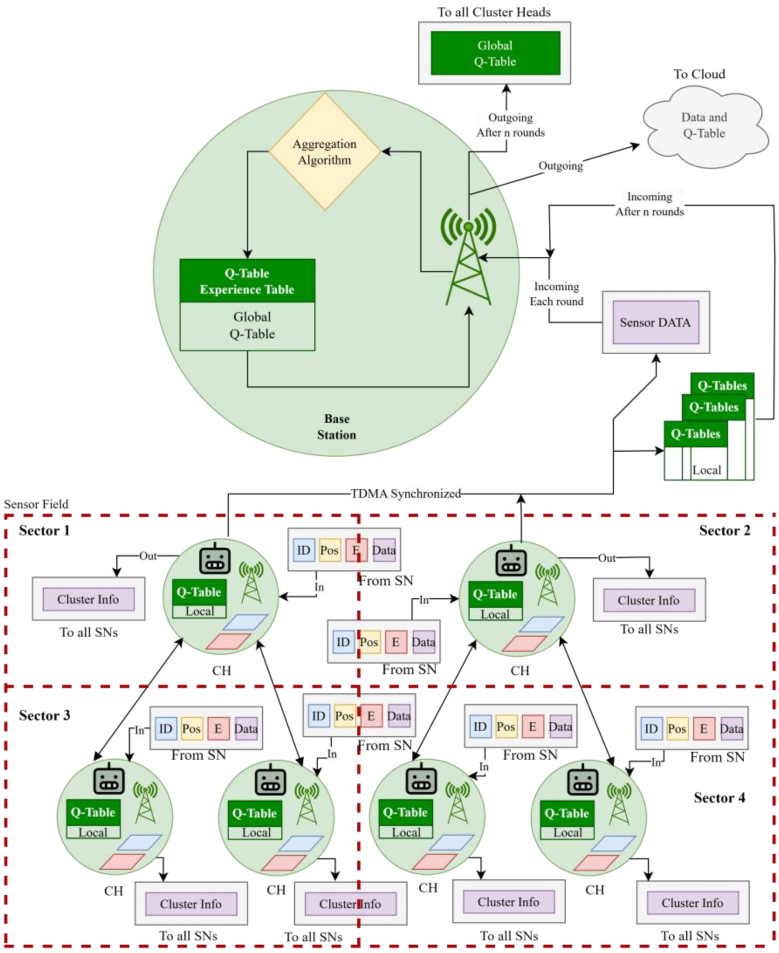


Figure 4.6.2 Federated RLHC Deployment

By integrating DEECP based candidate selection, spatial optimization through EEKA, compact clustering via K-Means, adaptive reinforcement learning and realistic communication modeling, the proposed pipeline forms a closed feedback loop capable of continuous optimization. The centralized model offers strong global optimization capability, whereas the federated learning configuration achieves superior robustness, scalability and long-term energy sustainability, making it particularly suitable for safety critical underground monitoring applications.

Chapter 5

Experimental Setup

The experimental setup defines the simulation environment, network configuration, energy model, learning parameters and evaluation metrics used to analyze the performance of the proposed framework. The objective of this setup is to ensure fair comparison between baseline clustering protocols and the proposed reinforcement learning–based hybrid clustering approach under identical operating conditions.

Each protocol is executed over multiple simulation rounds until termination conditions are reached. Randomness is controlled using consistent initialization to maintain reproducibility. During execution, statistics such as alive nodes, energy consumption, cluster stability and packet delivery performance are recorded at every round. The collected results are subsequently analyzed and compared in the Analysis and Discussion section.

Table 5.1 Network Parameters

Network Parameters	Value
WSN area size	1000m x 1000m
Initial energy	2 Joules
Number of Sensor Nodes	150
Packet size	128 bytes (1024 bits)
Data aggregation energy	5n J/bit/signal
Transmit energy	50n J/bit
Receive energy	50n J/bit
Free space loss	10n J/bit
Multipath loss	0.0013p J/bit
Base Station Mobility	Fixed
Network Type	Heterogeneous WSN
Simulation Rounds	1000-4000

Table 5.2 Node and Sink Model

Node and Sink Model	Value
Node Mobility	Random Walk
Maximum Node Speed	1 m/s
Sectorization	3 × 3 grid (9 sectors)
Operating Frequency	865 MHz (LoRa)
Channel Type	Composite channel

Chapter 6

Analysis and Discussions

The performance of the proposed RLHC framework is analyzed in terms of network lifetime, energy efficiency, clustering stability, packet delivery performance and adaptability under heterogeneous wireless sensor network conditions. The objective of this analysis is to evaluate how the integration of optimization algorithms and reinforcement learning improves overall network sustainability compared to conventional clustering approaches.

6.1 Sector-Wise Survival Analysis

Sector-wise survival analysis evaluates how uniformly nodes remain operational across different regions of the sensing field during network execution. The monitoring area is logically divided into multiple sectors based on spatial coordinates or angular partitioning around the base station. This analysis helps identify energy imbalance, hotspot formation and uneven cluster head distribution.

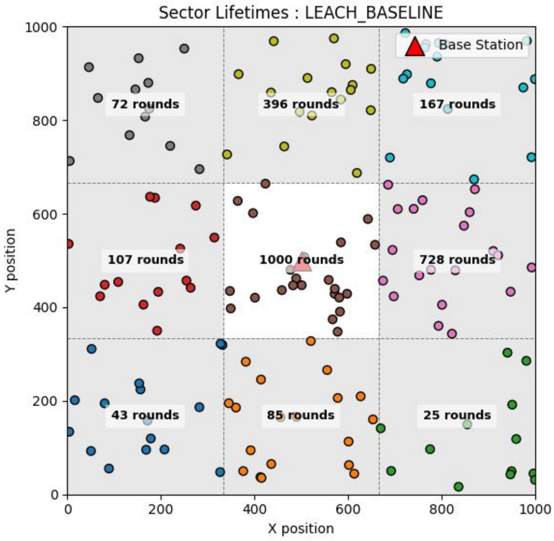


Fig 6.1 Sector-wise survival in LEACH

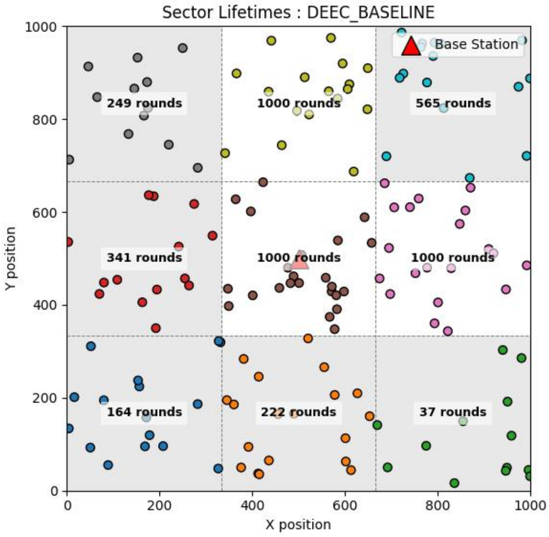


Fig 6.2 Sector-wise survival in DEEC

In the LEACH protocol, CH selection follows a probabilistic rotation mechanism without considering residual energy or node heterogeneity. Each node independently elects itself as a cluster head based on a predefined threshold probability. Here only 1 out of 9 sector survived.

Whereas DEEC improves sector survivability by incorporating residual energy awareness into cluster head election. CH probability is proportional to node energy relative to network average energy. High energy nodes assume leadership roles more frequently, reducing overload on weaker nodes. Here only 3 out of 9 sector survived.

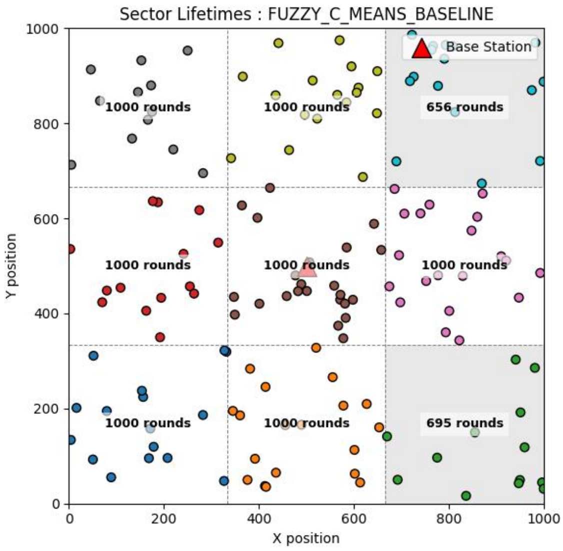


Fig 6.3 Sector-wise survival in FCM

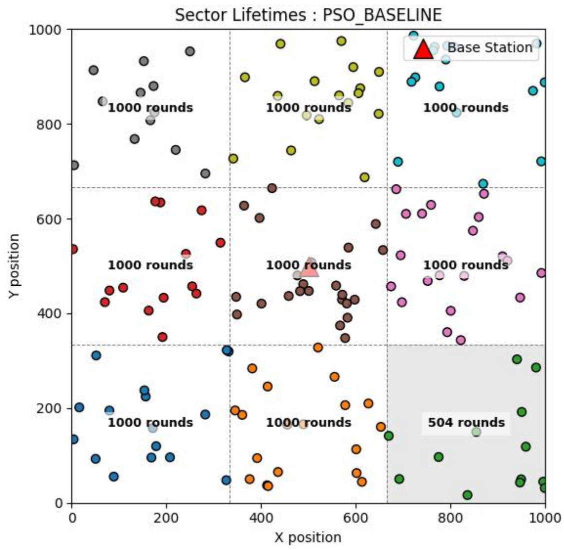


Fig 6.4 Sector-wise survival in PSO

In the FCM clustering approach, cluster formation is performed using fuzzy membership functions based primarily on spatial proximity between sensor nodes and cluster centroid. Nodes are assigned to clusters with varying degrees of membership, which improves cluster compactness and reduces intra cluster communication distance. However, CH positioning does not explicitly incorporate residual energy awareness or long-term communication load balancing. As a result CH located in dense sectors experience higher aggregation and forwarding responsibilities. Continuous energy drain in such regions leads to uneven node depletion across the sensing field. Consequently, several sectors experience premature collapse due to localized energy exhaustion. Here 7 out of 9 sectors survived.

In the PSO based clustering protocol, CH selection and positioning are optimized using swarm intelligence to minimize communication distance while maximizing residual energy utilization. PSO searches for globally optimal cluster configurations by considering both spatial distribution and energy conditions of nodes. This adaptive optimization prevents repeated cluster head selection within energy stressed regions and distributes communication load more uniformly across sectors. As a result, hotspot formation is reduced and energy consumption becomes balanced throughout the network. Sector degradation occurs gradually rather than abruptly, maintaining sensing coverage for a longer duration. Here 8 out of 9 sectors survived.

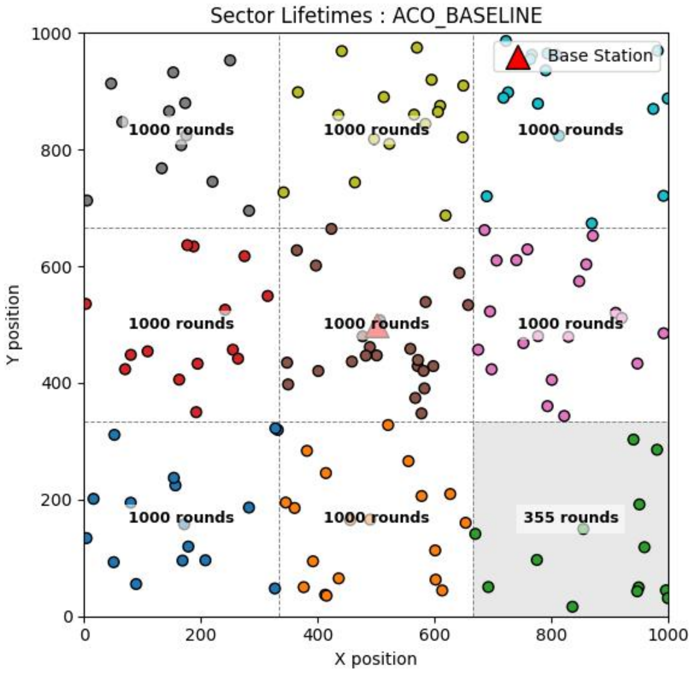


Fig 6.5 Sector-wise survival in ACO

In the ACO based clustering protocol, CH selection and routing decisions are guided by pheromone updating mechanisms and heuristic information such as communication distance and residual energy. During early network operation, ACO distributes cluster heads relatively well across the sensing field, reducing excessive communication distance within clusters. However, pheromone accumulation tends to favor frequently successful paths and nodes located in advantageous positions, such as centrally located or relay-heavy sectors. Over successive rounds, these regions may experience repeated selection for aggregation or forwarding tasks, increasing their energy depletion rate.

Although ACO improves routing efficiency and avoids completely random cluster formation, energy imbalance may still develop in sectors responsible for multi-hop forwarding toward the base station. Consequently, some sectors experience earlier node death compared to fully energy-aware optimization schemes. Nevertheless, sector degradation occurs more gradually than probabilistic clustering approaches due to adaptive path exploration.

As a result, sensing coverage is partially preserved across the monitoring area for longer duration compared to simpler protocols. Here 8 out of 9 sectors survived.

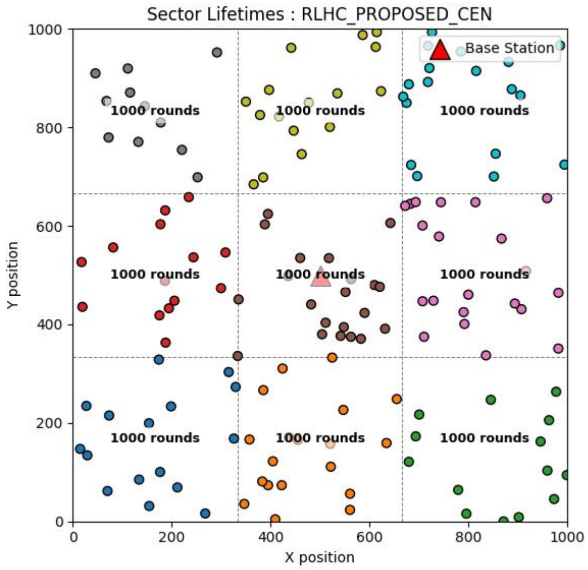


Fig 6.6 Sector-wise survival in RLHC

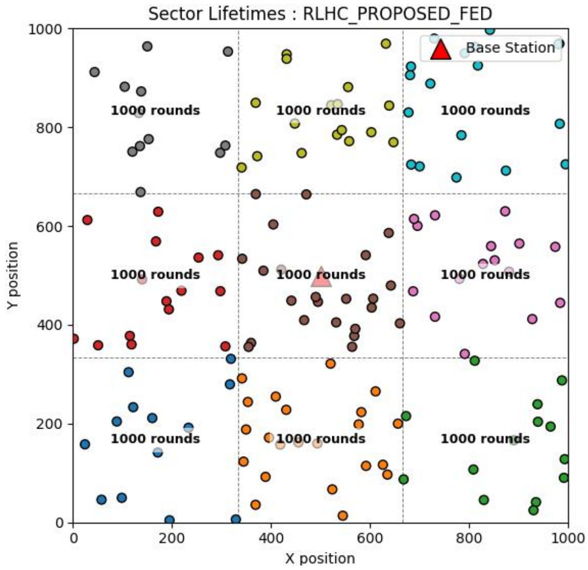


Fig 6.7 Sector-wise survival in RLHC

During early simulation rounds, centralized learning rapidly converges toward optimal decisions because the agent possesses full visibility of node residual energy, cluster load and communication reliability. Cluster heads are selected from energy-rich regions while avoiding excessive concentration within a single sector. This results in uniform cluster formation and reduced early energy depletion. As a result, sensing coverage remains operational in the majority of regions for an extended duration. Here 9 out of 9 sectors survived.

During early operation, local agents adapt rapidly to sector-specific conditions. Since learning occurs near the data source, communication overhead associated with frequent global state transmission is significantly reduced. Energy consumption remains lower compared to centralized learning during coordination phases. This distributed intelligence prevents relay-heavy sectors from becoming permanent communication bottlenecks. Energy utilization becomes highly balanced and node survival curves across sectors remain closely aligned. In later rounds, federated RLHC shows superior robustness. Even if some sectors begin to degrade, other sectors continue learning independently, maintaining sensing continuity. Sector collapse occurs slowly and uniformly across the network. Here 9 out of 9 sectors survived.

6.2 Network Lifetime Analysis in Terms of Dead Nodes

49 Network lifetime is a critical performance metric in wireless sensor networks and is commonly evaluated by observing the progression of node deaths across simulation rounds. Dead node analysis provides insight into energy dissipation behaviour, load balancing efficiency and clustering stability of different protocols. A node is considered dead when its residual energy becomes zero and it can no longer participate in sensing, aggregation, or communication activities.

The lifetime of the network is typically characterized using three major milestones:

i. First Node Death (FND) — indicates stability period.
ii. Half Node Death (HND) — represents network operational reliability.
iii. Last Node Death (LND) — determines overall lifetime termination.

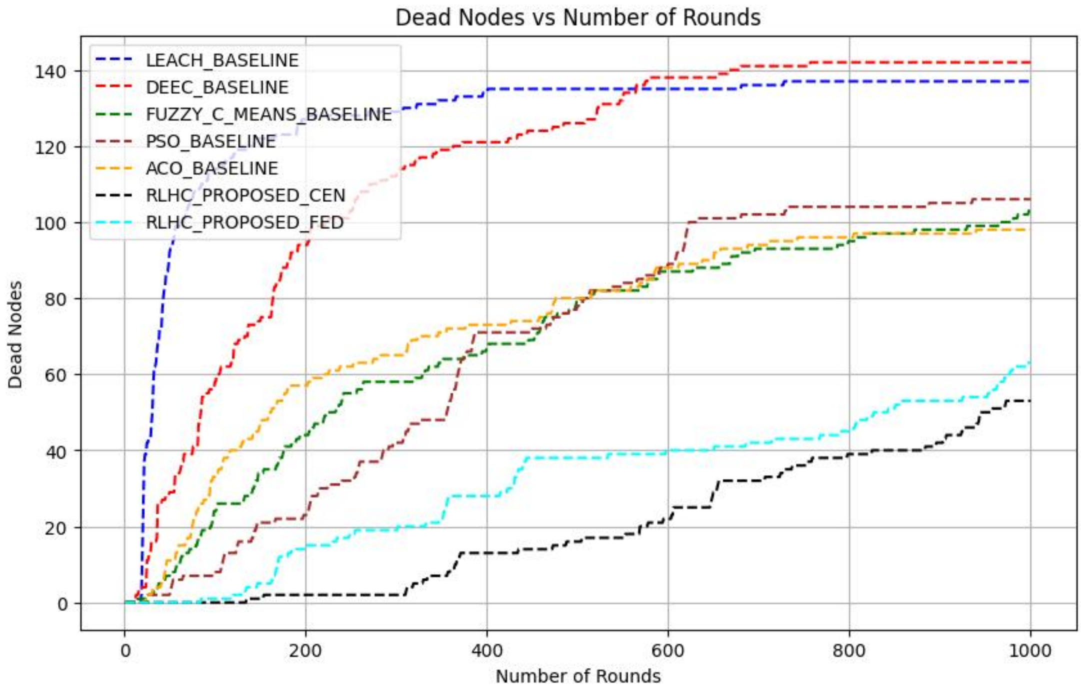


Figure 6.2.1 Network lifetime in terms of node death

28 Network lifetime performance is evaluated by analyzing the number of dead nodes at different simulation durations. Dead nodes indicate energy exhaustion and directly reflect the stability and sustainability of clustering protocols.

At 1000 simulation rounds, the LEACH protocol shows the highest energy depletion with 138 dead nodes out of 150, indicating rapid network degradation caused by random cluster head selection. Similarly, DEEC records 142 dead nodes, showing that although energy awareness improves sector survival, relay burden and heterogeneous traffic still lead to significant node loss. Optimization-based approaches demonstrate improved survivability. FCM results in 107 dead nodes, while PSO reduces node deaths further to 108 nodes through optimized cluster formation. ACO achieves better performance with 94 dead nodes, benefiting from

adaptive routing and pheromone-based path optimization. The proposed RLHC approaches significantly outperform conventional protocols. The RLHC Centralized model records only 55 dead nodes, representing the lowest energy depletion due to global decision making and balanced cluster head rotation. The RLHC Federated model shows 62 dead nodes, slightly higher than centralized learning but still maintaining strong survivability due to reduced communication overhead and localized adaptation.
At extended operation of 4000 simulation rounds, network aging becomes more evident. RLHC Centralized experiences severe degradation with 140 dead nodes, leaving only one sector operational due to continuous coordination overhead and relay pressure near the sink. In contrast, the RLHC Federated model maintains improved survivability with 130 dead nodes and three sectors still active, demonstrating stronger long-term resilience.

Overall, dead node progression confirms that reinforcement learning–based adaptive clustering significantly extends operational lifetime compared to traditional and optimization-based protocols. While centralized learning achieves strong short-term performance, federated reinforcement learning provides superior long-term sustainability under prolonged network operation.

6.3 Energy Consumption Analysis

Energy consumption directly reflects the efficiency of clustering and communication mechanisms in wireless sensor networks. At 1000 simulation rounds, LEACH and DEEC show the highest energy usage with 297 J and 295 J, respectively, due to random or relay-heavy cluster head selection leading to rapid energy depletion.

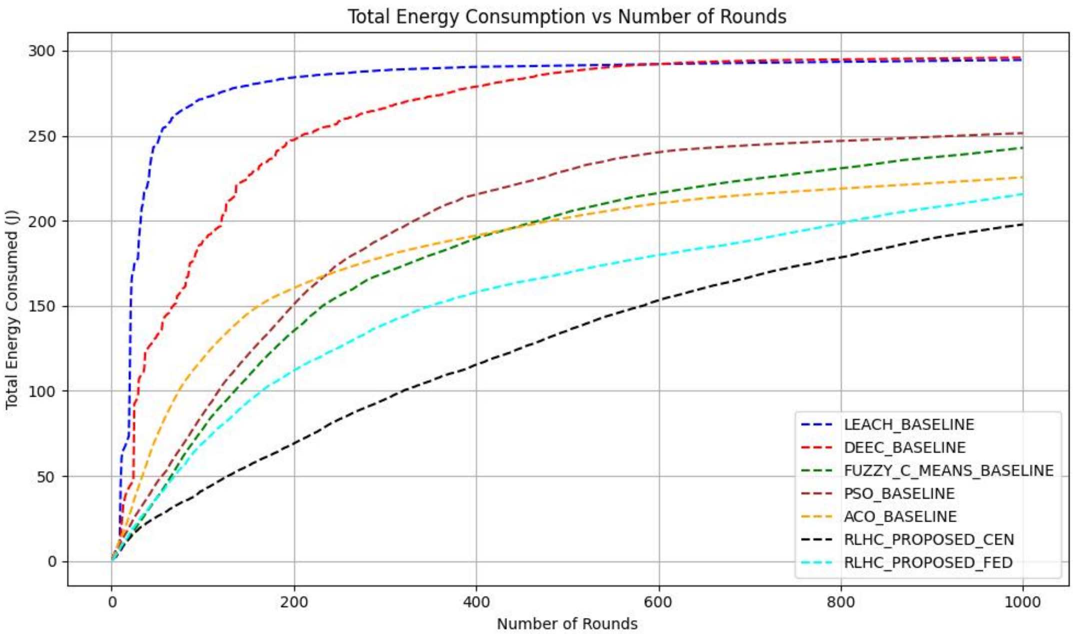


Figure 6.3.1 Total Energy Consumption

Optimization-based approaches demonstrate better performance. FCM consumes 240 J, PSO uses 251 J and ACO further reduces consumption to 220 J through optimized routing and balanced communication load. The proposed RLHC approaches achieve the best efficiency. RLHC Centralized consumes only 197 J, showing maximum short-term energy savings due to global optimization, while RLHC Federated consumes 214 J, maintaining balanced energy utilization with reduced communication overhead.
At 4000 simulation rounds, RLHC Centralized energy consumption increases to 297 J due to continuous coordination overhead, whereas RLHC Federated consumes 286 J, demonstrating better long-term energy sustainability.

Overall, RLHC methods significantly reduce energy consumption compared to conventional and optimization-based protocols, with federated learning providing superior long-term efficiency.

6.4 Throughput Analysis

the total throughput performance of different clustering protocols in terms of successful data transmission to the base station. Throughput represents the amount of useful data delivered per unit time and reflects communication efficiency, cluster stability and routing reliability.

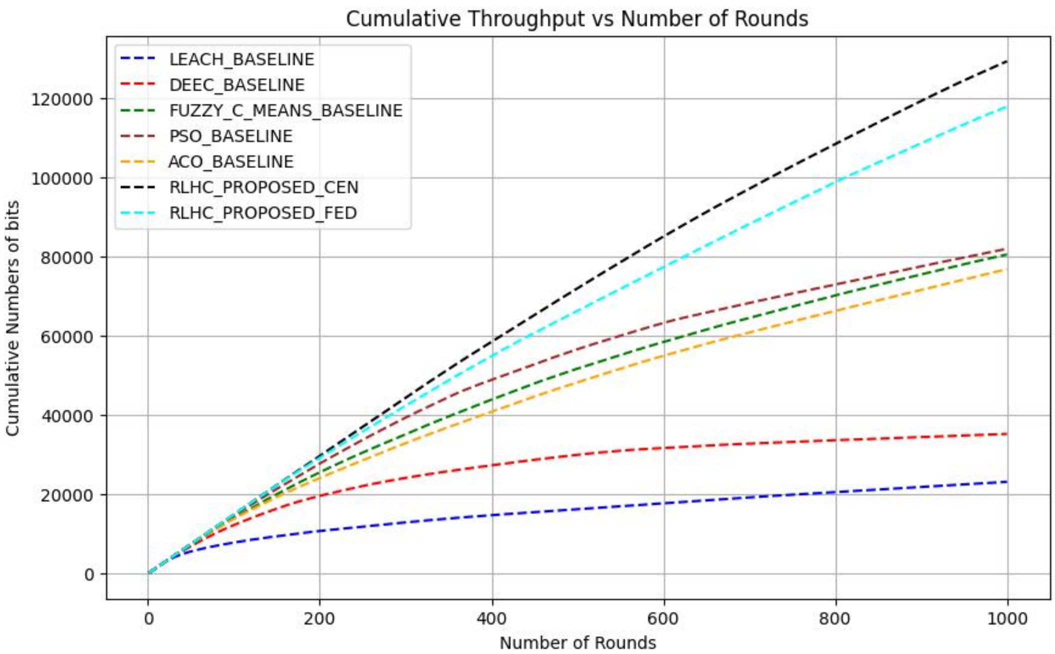


Figure 6.4 Cumulative Throughput

At 1000 simulation rounds, LEACH achieves the lowest throughput of 22 kbps due to random cluster head selection and frequent packet loss caused by early node failures. DEEC improves performance to 38 kbps by incorporating residual energy awareness, however, relay congestion limits further improvement.
Among optimization-based approaches, FCM achieves 80 kbps, while PSO records 81 kbps, benefiting from compact clustering and optimized cluster head placement.

ACO achieves 77 kbps, where pheromone-guided routing improves link selection but repeated relay usage slightly affects performance. The proposed RLHC approaches demonstrate significant improvement. The RLHC Centralized model achieves the highest throughput of 133 kbps, supported by global decision making and efficient load balancing across clusters. The RLHC Federated model achieves 118 kbps, slightly lower than centralized learning but maintaining stable communication through localized adaptive optimization.
At extended operation of 4000 simulation rounds, throughput increases and RLHC Centralized achieves 260 kbps, while RLHC Federated maintains 250 kbps, demonstrating sustained communication reliability even during late network stages.

Overall, reinforcement learning–based clustering significantly enhances throughput by improving cluster stability, reducing packet loss and optimizing routing decisions compared to conventional and optimization-based protocols

6.5 Result Summary

Table 6.5.1 Comparison table for 1000 rounds of simulation for 150 nodes

Simulation for 1000 rounds					
Methods	Sector Survived	No.dead nodes	Energy Usage (J)	Throughput (kbps)	Avg PDR (%)
LEACH	1/9	138/150	297/300	22	99.74
DEEC	3/9	142/150	295/300	38	99.71
PSO	8/9	108/150	251/300	81	99.74
ACO	8/9	94/150	220/300	77	99.86
FCM	7/9	107/150	240/300	80	99.52
RLHC Centralized	9/9	55/150	197/300	133	99.92
RLHC Federated	9/9	62/150	214/300	118	99.95

Table 6.5.2 Comparison table for 4000 rounds of simulation for 150 nodes

Simulation for 4000 rounds					
Methods	Sector Survived	No.dead nodes	Energy Usage (J)	Throughput (kbps)	Avg PDR (%)
RLHC Centralized	1/9	140/150	297/300	260	99.87
RLHC Federated	3/9	130/150	286/300	250	99.97

The results show that LEACH and DEEC experience faster energy depletion and early node deaths due to inefficient load balancing. FCM, PSO and ACO improve clustering efficiency and network stability. The proposed RLHC approaches achieve the best performance, where RLHC Centralized provides highest throughput and short-term efficiency, while RLHC Federated ensures better sector survivability and longer network lifetime through balanced energy consumption and adaptive learning.

29

Chapter 7

Conclusion and Future Scope

This research proposed a Reinforcement Learning based Hybrid Clustering (RLHC) framework to address major challenges related to energy efficiency, network lifetime, sector survivability and reliable communication in heterogeneous wireless sensor networks. Conventional clustering protocols such as LEACH and DEEC rely on probabilistic or residual energy–based cluster head selection, which often leads to hotspot formation, uneven load distribution and premature node failures in large-scale and relay-intensive environments. Optimization-based approaches including FCM, PSO and ACO improve clustering compactness and routing efficiency using intelligent optimization techniques. However, their static decision strategies limit adaptability under dynamic network conditions such as residual energy variation, link degradation and sector-level traffic imbalance.

50

To overcome these limitations, the proposed RLHC framework integrates energy-aware candidate selection, spatial optimization, compact clustering, reinforcement learning–based adaptive decision making and communication management within a unified end-to-end pipeline. Reinforcement learning enables continuous interaction with the network environment, allowing adaptive cluster head selection and routing optimization based on network feedback.

7.1 Conclusion

Simulation results for 1000 rounds demonstrated clear performance improvements. LEACH and DEEC showed poor survivability with only 1/9 and 3/9 active sectors, respectively, along with high dead node counts (138 and 142 nodes) and maximum energy consumption (297 J and 295 J). Optimization approaches improved performance, where PSO and ACO maintained 8/9 sectors, FCM sustained 7/9 sectors and reduced dead nodes to the range of 94–108 nodes with energy usage between 220–251 J.

In contrast, RLHC achieved the best overall performance. RLHC Centralized maintained 9/9 sector survivability, reduced dead nodes to 55/150, minimized energy consumption to 197 J and achieved the highest throughput of 133 kbps with 99.92% PDR. RLHC Federated also sustained 9/9 sectors, with 62 dead nodes, 214 J energy usage, 118 kbps throughput and the highest reliability of 99.95% PDR, demonstrating balanced long-term stability.

Extended simulations for 4000 rounds further highlighted long-term behaviour. RLHC Centralized retained only 1/9 surviving sector with 140 dead nodes and 297 J energy usage, indicating faster exhaustion under prolonged operation despite high throughput (260 kbps). In comparison, RLHC Federated maintained 3/9 sectors, fewer dead nodes (130/150), lower energy consumption (286 J), sustained high throughput (250 kbps) and superior reliability (99.97% PDR), confirming improved scalability and long-duration robustness.

Overall, the RLHC framework successfully combines deterministic optimization with adaptive reinforcement learning intelligence to achieve balanced energy utilization, improved sector survivability, extended network lifetime and enhanced communication reliability. The federated learning configuration particularly demonstrates strong suitability for large-scale, long-duration deployments such as underground mining monitoring, smart infrastructure surveillance and industrial IoT environments.

7.2 Future Scope

37 Future work can focus on improving the scalability of RLHC for large-scale wireless sensor networks through distributed and multi-agent reinforcement learning to reduce coordination overhead. Integration of deep reinforcement learning and graph-based learning models may further enhance adaptive cluster head selection and routing decisions under dynamic network conditions. Energy harvesting–aware clustering can also be incorporated to extend operational lifetime in long-term monitoring applications.

Real-world validation using IoT edge hardware and deployment in applications such as underground mining monitoring, smart infrastructure and industrial environments remains an important direction. Additionally, future research may include security-aware learning mechanisms for fault tolerance and integration with emerging technologies such as UAV assisted data collection and next-generation communication networks to enable fully autonomous intelligent network management.